Q1.

A group of friends want to prepare a meal. They start preparing the meal at 6:30pm

Activities to prepare the meal are shown in **Figure 1** below.

Figure 1

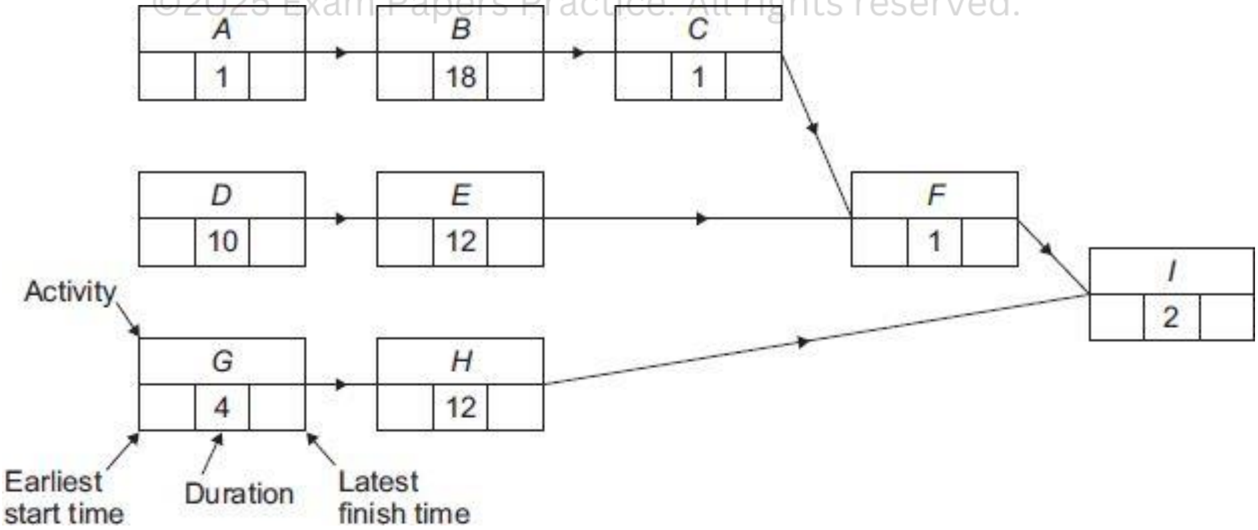
Label	Activity	Duration (mins)	Immediate predecessors
A	Weigh rice	1	–
B	Cook rice	18	*A*
C	Drain rice	1	*B*
D	Chop vegetables	10	–
E	Fry vegetables	12	
F	Combine fried vegetables and drained rice	1	
G	Prepare sauce ingredients	4	–
H	Boil sauce	12	
I	Serve meal on plates	2	

(a) (i) Use **Figure 2** shown below to complete **Figure 1** above.

(1)

(ii) Complete **Figure 2** showing the earliest start time and latest finishing time for each activity.

Figure 2



(2)

(b) (i) State the activity which must be started first so that the meal is served in the shortest possible time.

Fully justify your answer.

(2)

(ii) Determine the earliest possible time at which the preparation of the meal can be completed.

(1)

(c) The group of friends want to cook spring rolls so that they are served at the same time as the rest of the meal. This requires the additional activities shown in **Figure 3**.

Figure 3

Label	Activity	Duration	Immediate predecessors
J	Switch on and heat oven		–
K	Put spring rolls in oven and cook		
L	Transfer spring rolls to serving dish		

It takes 15 seconds to switch on the oven.

The oven must be allowed to heat up for 10 minutes before the spring rolls are put in the oven.

It takes 15 seconds to put the spring rolls in the oven.

The spring rolls must cook in the hot oven for 8 minutes.

It takes 30 seconds to transfer the spring rolls to a serving dish.

(i) Complete **Figure 3** above.

(1)

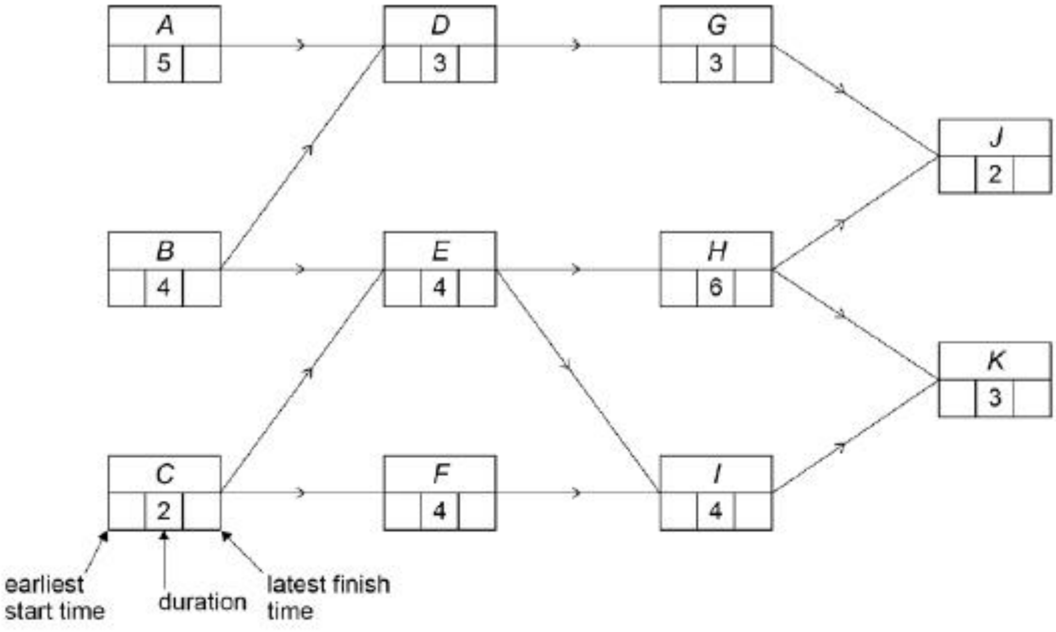
(ii) Determine the latest time at which the oven can be switched on in order for the spring rolls to be served at the same time as the rest of the meal.

(2)

(Total 9 marks)

Q2.

Deva Construction Ltd undertakes a small building project. The activity network for this project is shown below, where each activity's duration is given in hours.



(a) Complete the activity network for the building project.

(2)

(b) Deva Construction Ltd is able to reduce the duration of a single activity to 1 hour by using specialist equipment.

State, with a reason, which activity should have its duration reduced to 1 hour in order to minimise the completion time for the building project.

(3)

(c) State one limitation in the building project used by Deva Construction Ltd.

Explain how this limitation affects the project.

(2)

(Total 7 marks)

Q3.

Optical fibre broadband cables are being installed between 5 neighbouring villages.

The distance between each pair of villages in metres is shown in the table.

	Alvanley	**Dunham**	**Elton**	**Helsby**	**Ince**
Alvanley	-	2000	4000	750	5500
Dunham	2000	-	2500	2250	4000
Elton	4000	2500	-	3000	1250
Helsby	750	2250	3000	-	4250
Ince	5500	4000	1250	4250	-

The company installing the optical fibre broadband cables wishes to create a network connecting each of the 5 villages using the minimum possible length of cable.

Find the minimum length of cable required.

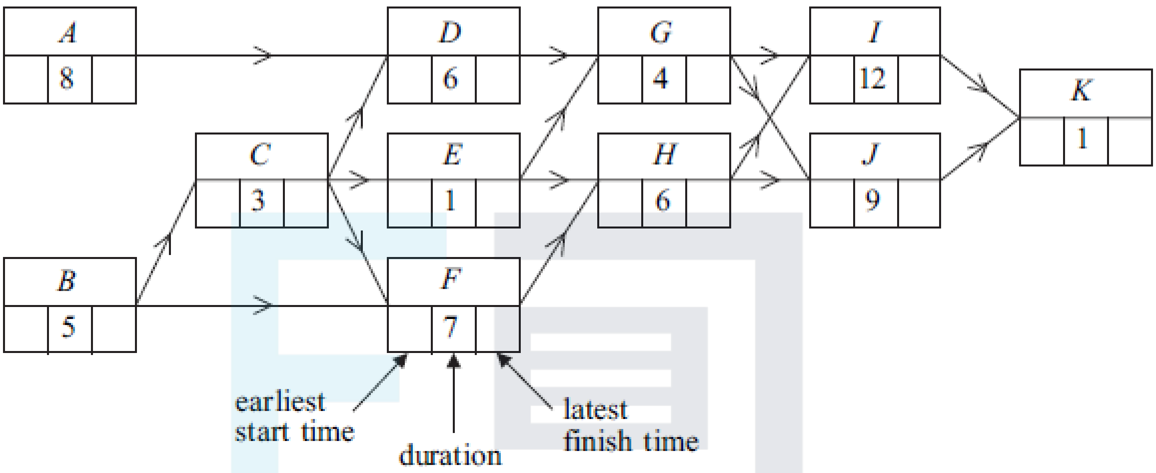
(Total 3 marks)

Q4.

Figure 1 below shows an activity diagram for a project. Each activity requires one worker. The duration required for each activity is given in hours.

(a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



(4)

(b) On **Figure 2** complete the precedence table.

Figure 2

Activity	Immediate predecessor(s)
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	
K	

(2)

(c) Find the critical path.

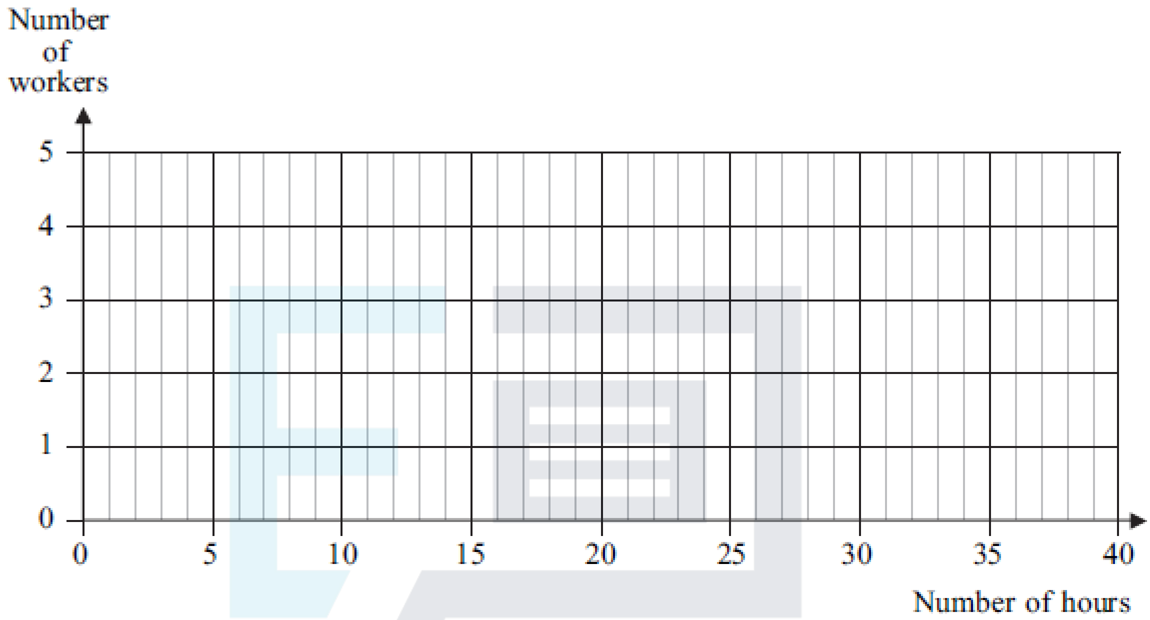
(1)

(d) Find the float time of activity E.

(1)

(e) Using **Figure 3**, draw a resource histogram to illustrate how the project can be completed in the minimum time, assuming that each activity is to start as early as possible.

Figure 3



(3)

(f) Given that there are two workers available for the project, find the minimum completion time for the project.

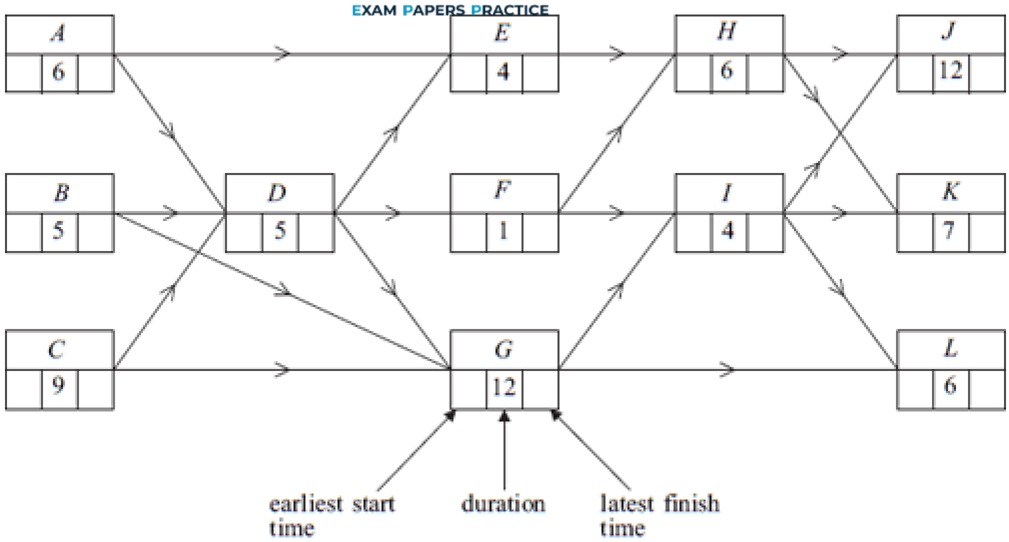
(1)

(g) Given that there is only one worker available for the project, find the minimum completion time for the project.

(1)
(Total 13 marks)

Q5.

The figure below shows an activity diagram for a project. The duration required for each activity is given in hours. The project is to be completed in the minimum time.



(a) Find the earliest start time and the latest finish time for each activity and insert their values on the figure.

(4)

(b) Find the critical path.

(1)

(c) Find the float time of activity E.

(1)

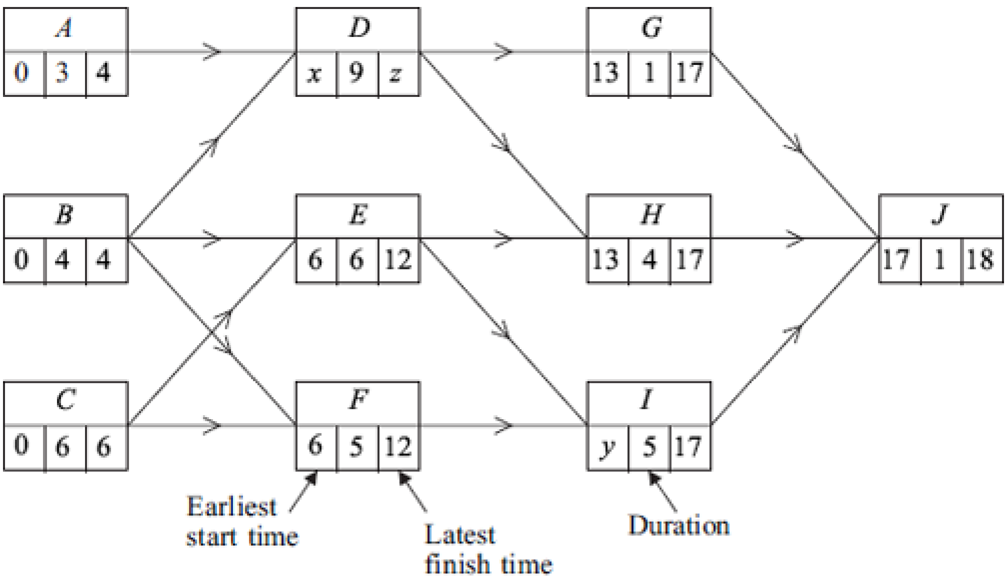
(d) Given that activities H and K will both overrun by 10 hours, find the new minimum completion time for the project.

(3)

(Total 9 marks)

Q6.

The diagram shows the activity network and the duration, in days, of each activity for a particular project. Some of the earliest start times and latest finish times are shown on the diagram.



(a) Find the values of the constants x, y and z.

(3)

(b) Find the critical paths.

(2)

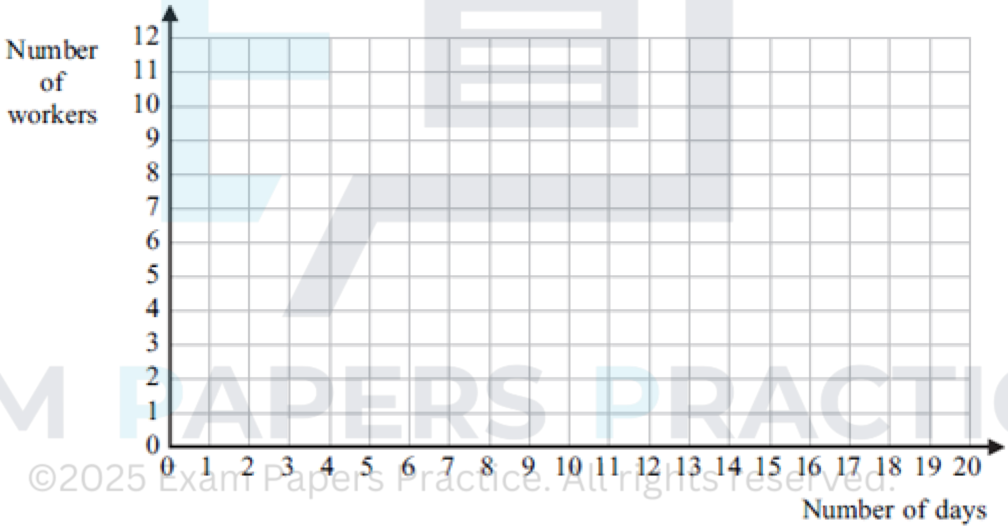
(c) Find the activity with the largest float and state the value of this float.

(2)

(d) The number of workers required for each activity is shown in the table.

Activity	A	B	C	D	E	F	G	H	I	J
Number of workers required	4	2	3	4	2	4	3	3	5	6

Given that each activity starts as **early** as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on the **figure** below, indicating clearly which activities are taking place at any given time.



(5)

(e) It is later discovered that there are only 9 workers available at any time. Use resource levelling to find the new earliest start time for activity J so that the project can be completed with the minimum extra time. State the minimum extra time required.

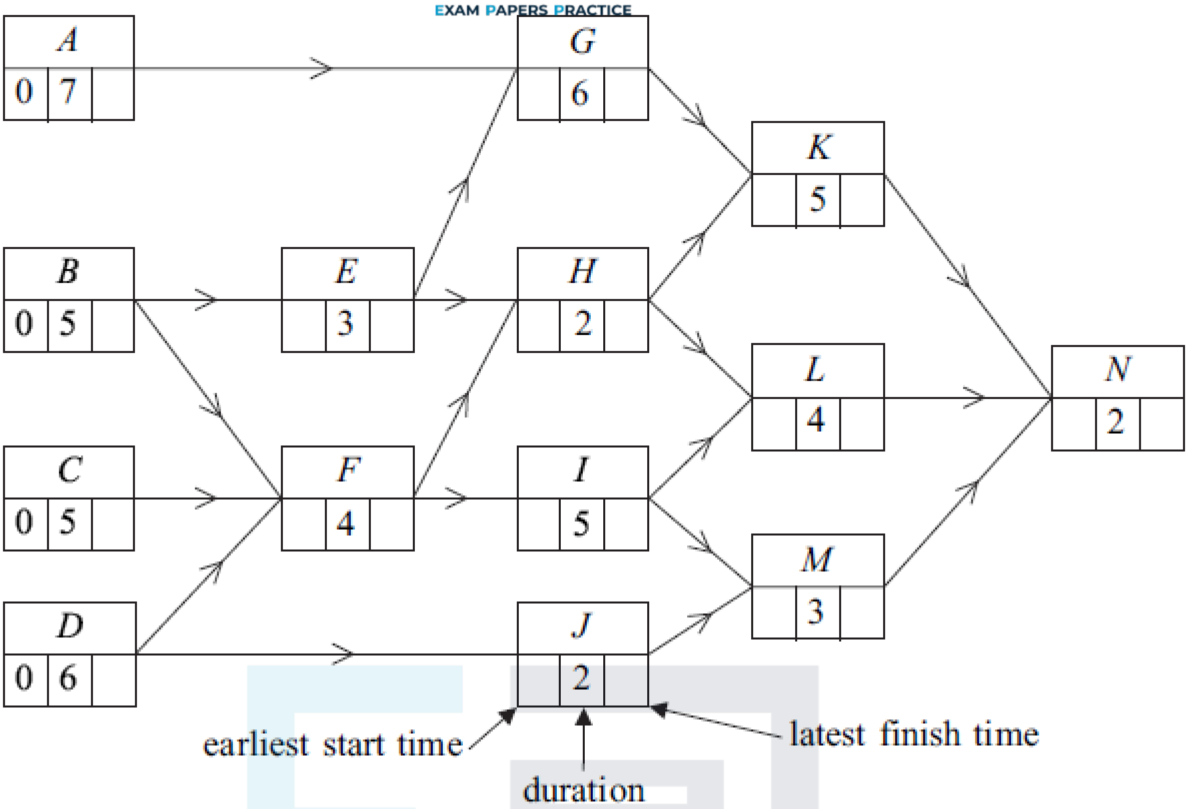
(2)

(Total 14 marks)

Q7.

Figure 1 below shows an activity diagram for a construction project. The time needed for each activity is given in days.

Figure 1



(a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

(4)

(b) Find the critical paths and state the minimum time for completion of the project.

Critical paths are ____

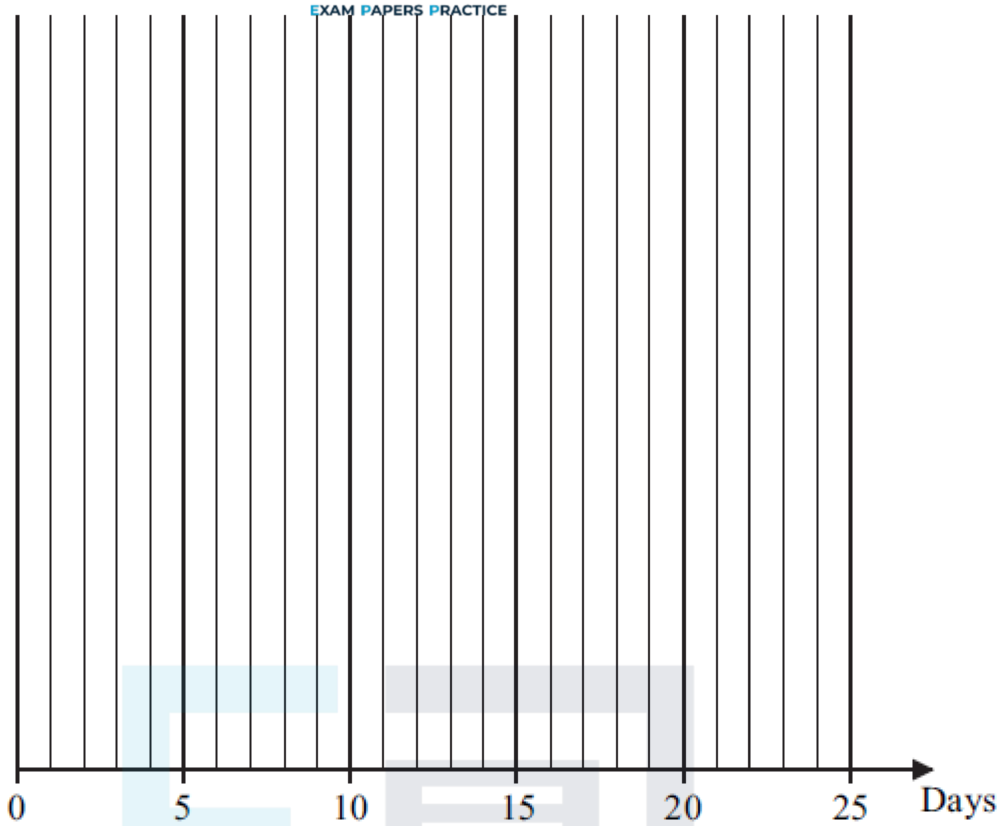
____

Minimum completion time is ______________ days.

(3)

(c) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as early as possible.

Figure 2



(5)

(d) Activity J takes longer than expected so that its duration is x days, where $x \geq 3$. Given that the minimum time for completion of the project is unchanged, find a further inequality relating to the maximum value of x.

(2)
(Total 14 marks)

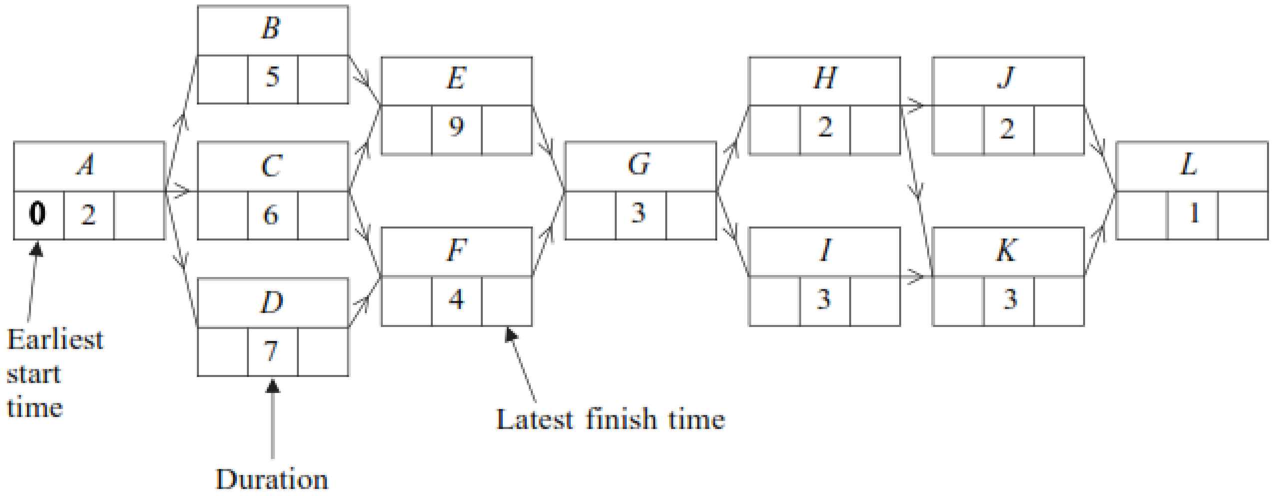
Q8.

A group of workers is involved in a decorating project. The table shows the activities involved. Each worker can perform any of the given activities.

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Duration (days)	2	5	6	7	9	4	3	2	3	2	3	1
Number of workers required	6	3	5	2	5	2	4	4	5	3	2	4

The activity network for the project is given in **Figure** 1 below.

Figure 1



(a) Find the earliest start time and the latest finish time for each activity, inserting their values on **Figure 1**.

(4)

(b) Hence find:

(i) the critical path;

(ii) the float time for activity D.

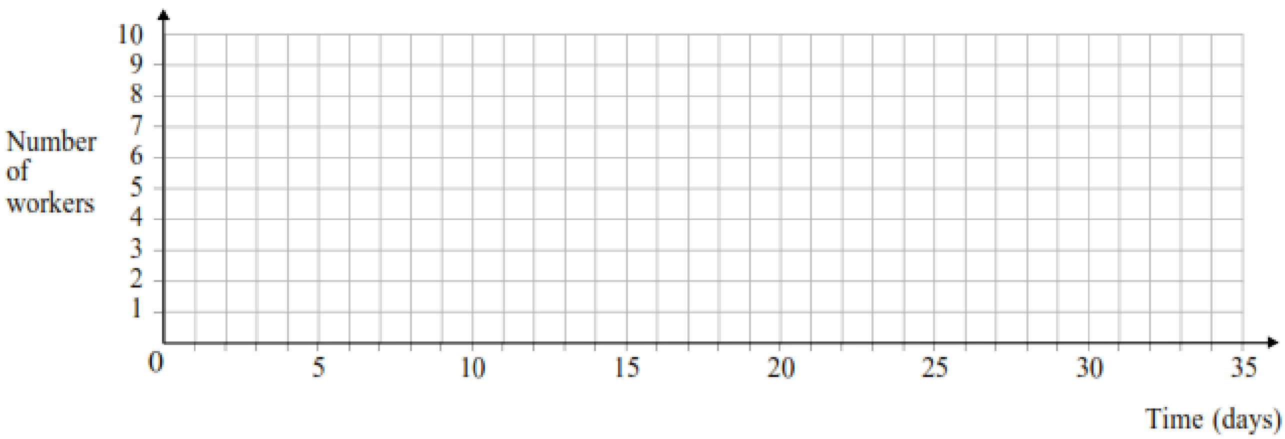
(3)

(i) The critical path is ____

(ii) The float time for activity D is ________________________________

(c) Given that each activity starts as early as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2** below, indicating clearly which activities are taking place at any given time.

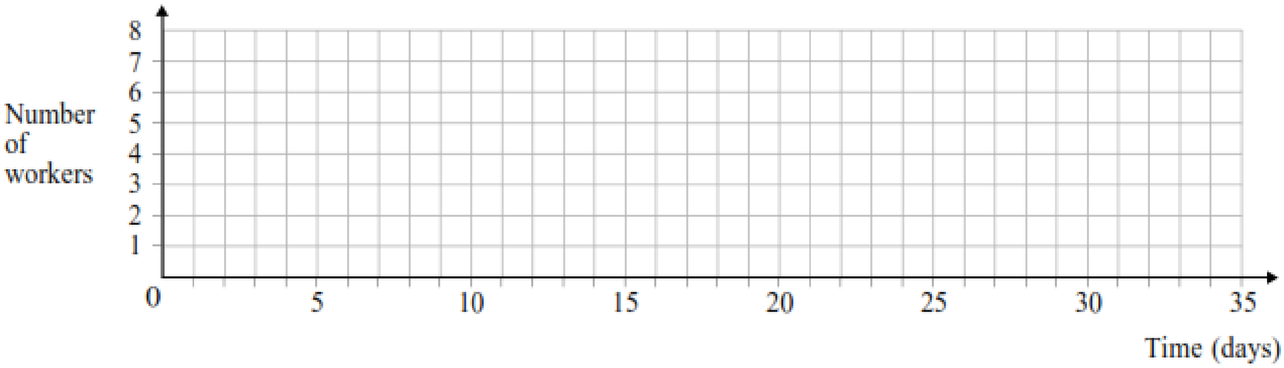
Figure 2



(4)

(d) It is later discovered that there are only 8 workers available at any time. Use resource levelling to construct a new resource histogram on **Figure 3** below, showing how the project can be completed with the minimum extra time. State the minimum extra time required.

Figure 3



The minimum extra time required is ________________________________

(3)

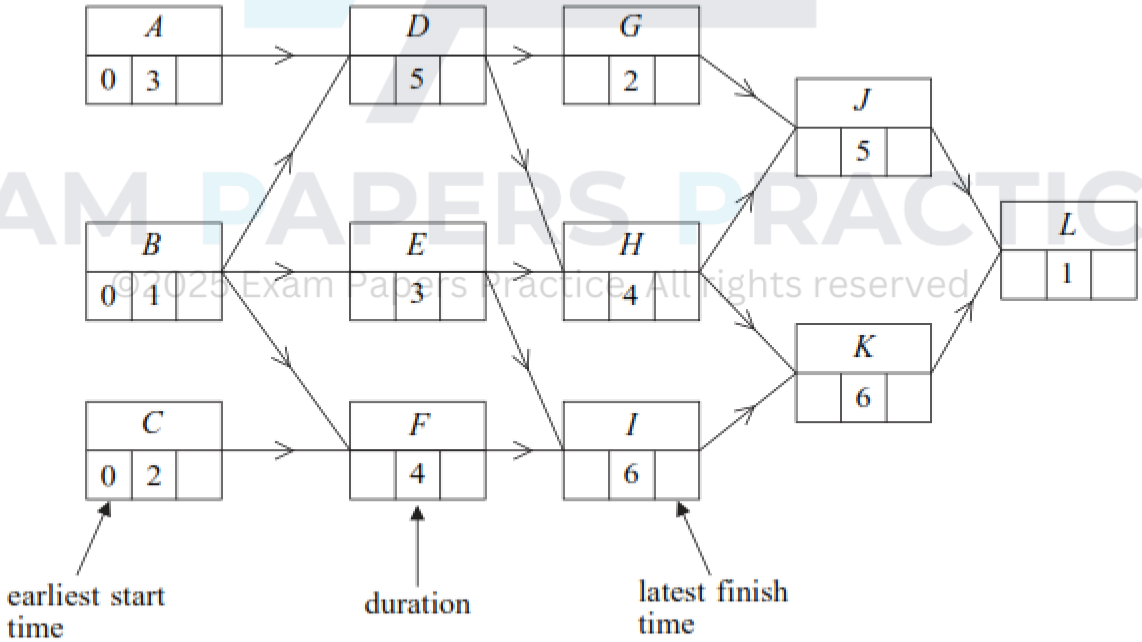
(Total 14 marks)

Q9.

Figure 1 below shows an activity diagram for a cleaning project. The duration of each activity is given in days.

(a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



(4)

(b) Find the critical paths and state the minimum time for completion of the project.

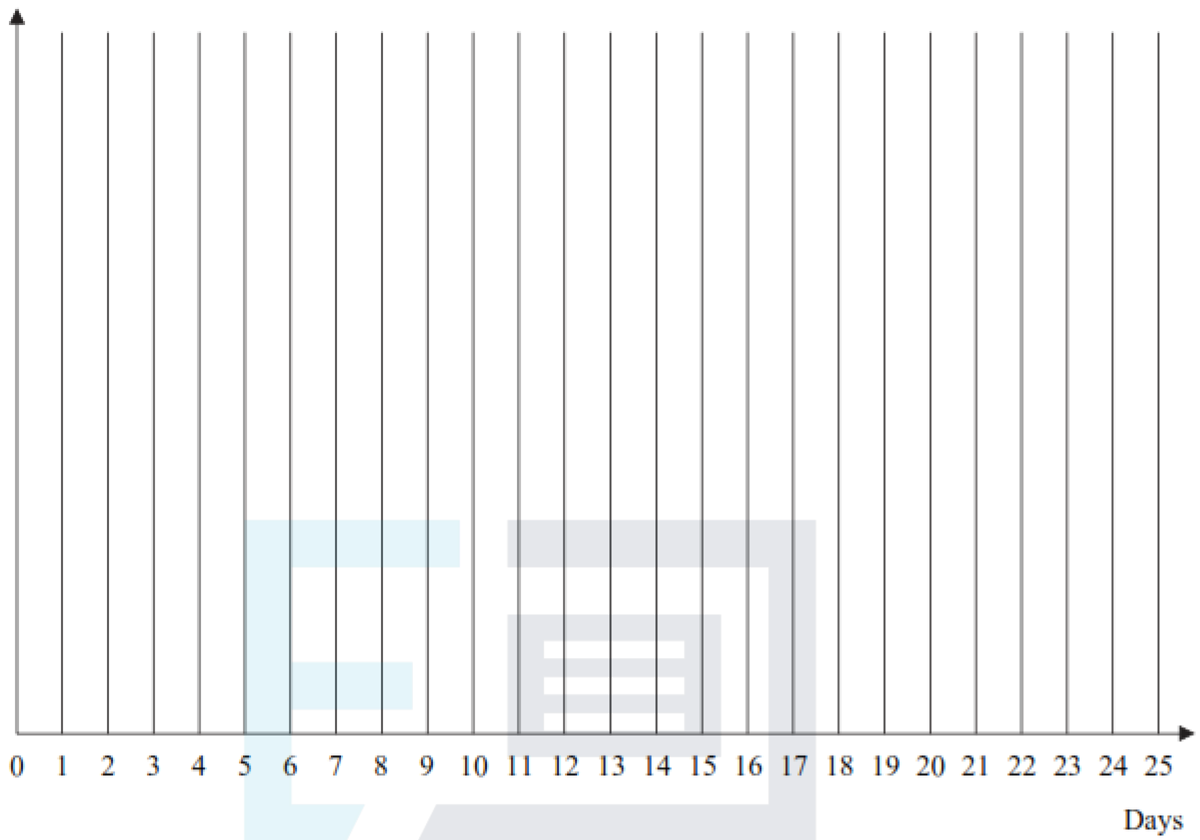
(3)

(c) Find the activity with the greatest float time and state the value of its float time.

(2)

(d) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as **late** as possible.

Figure 2



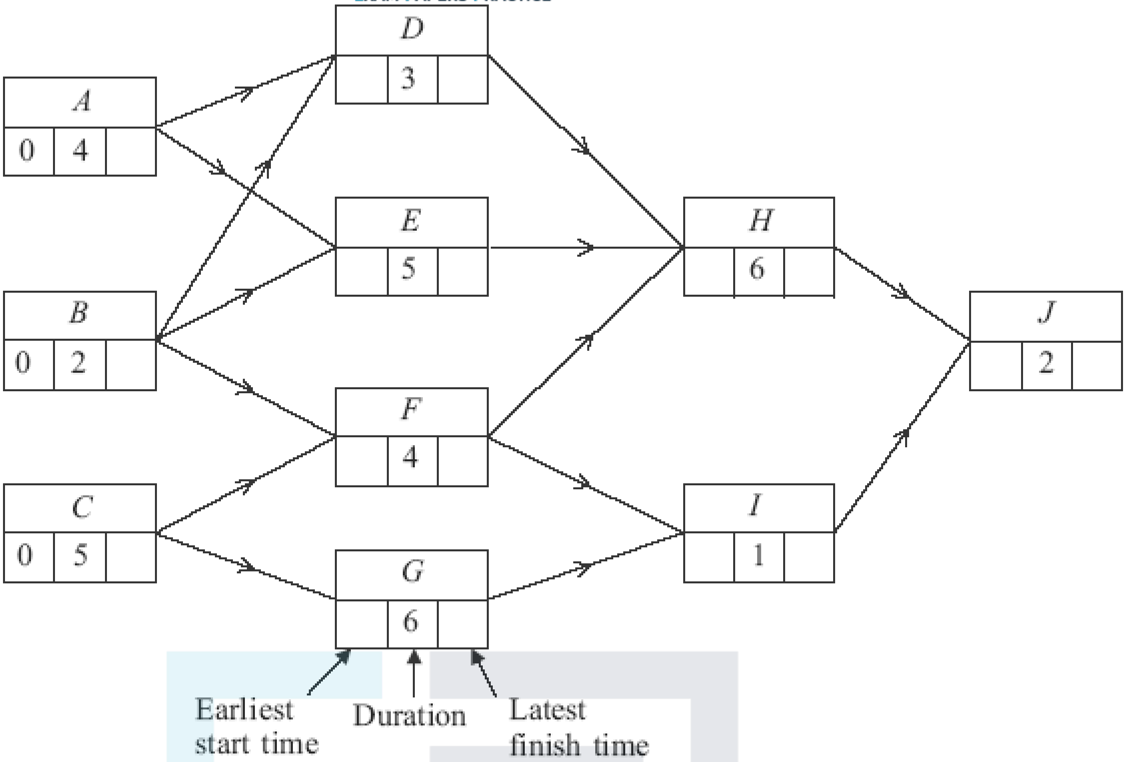
(4)

(Total 13 marks)

Q10.

Figure 1 shows the activity network and the duration, in days, of each activity for a particular project.

Figure 1



(a) On **Figure 1**:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Find the float for activity G.

(1)

(c) Find the critical paths and state the minimum time for completion.

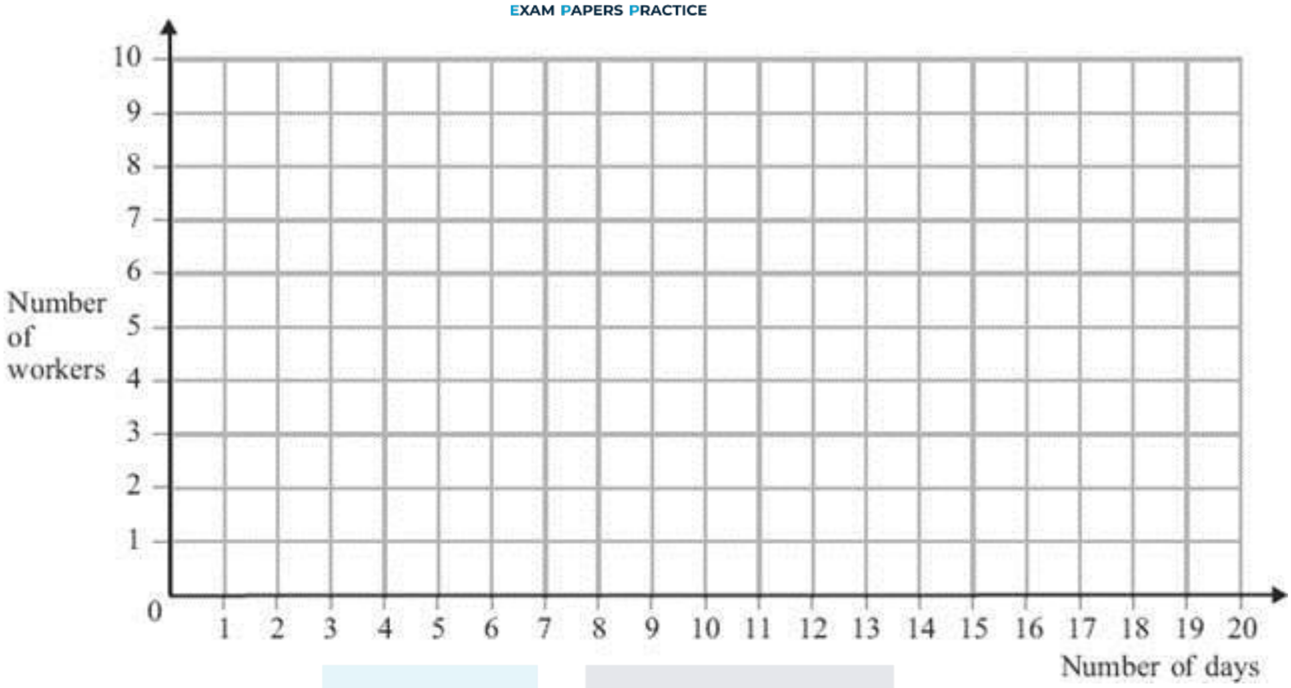
(3)

(d) The number of workers required for each activity is shown in the table.

Activity	A	B	C	D	E	F	G	H	I	J
Number of workers required	2	2	3	2	3	2	1	3	5	2

Given that each activity starts as **late** as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2**, indicating clearly which activities take place at any given time.

Figure 2

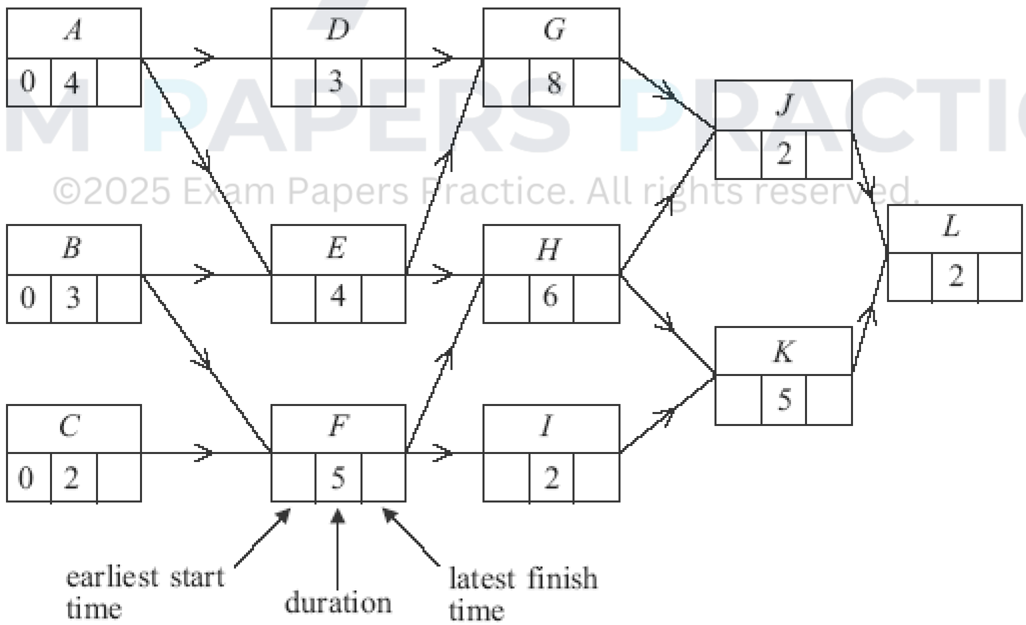


(5)
(Total 13 marks)

Q11.

Figure 1 below shows an activity diagram for a construction project. The time needed for each activity is given in days.

Figure 1



(a) Find the earliest start time and latest finish time for each activity and insert their values on **Figure 1**.

(4)

(b) Find the critical paths and state the minimum time for completion of the project.

Critical paths are ____

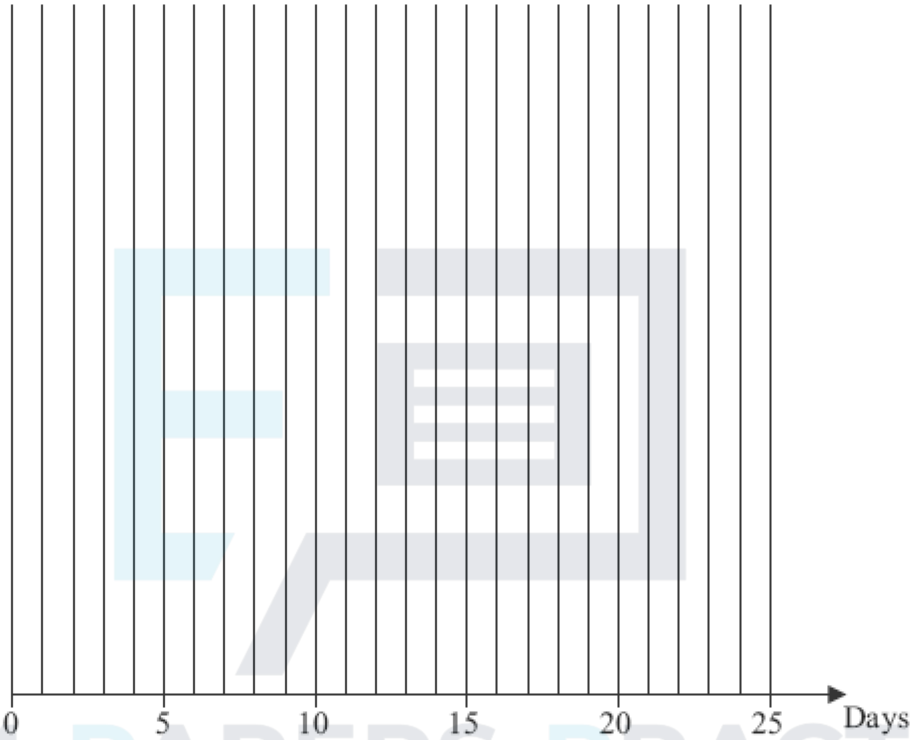
____

Minimum completion time is ____________________ days

(3)

(c) On **Figure 2** below, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as early as possible.

Figure 2



(3)

(d) A delay in supplies means that Activity I takes 9 days instead of 2.

(i) Determine the effect on the **earliest** possible starting times for activities K and L.

(2)

(ii) State the number of days by which the completion of the project is now delayed.

(1)

(Total 13 marks)

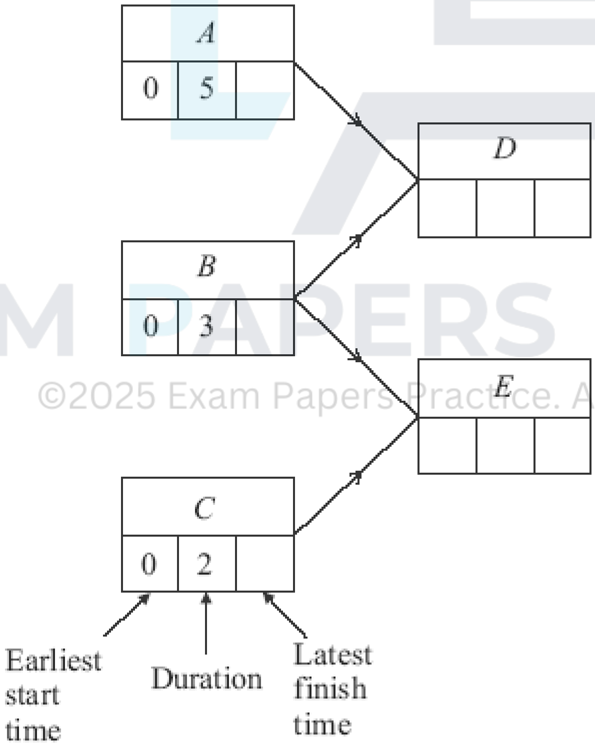
Q12.

A decorating project is to be undertaken. The table shows the activities involved.

Activity	**Immediate Predecessors**	**Duration (days)**
A	–	5

B	–	3
C	–	2
D	A, B	4
E	B, C	1
F	D	2
G	E	9
H	F, G	1
I	H	6
J	H	5
K	I, J	2

(a) Complete an activity network for the project below.



(3)

(b) On the diagram above, indicate:

(i) the earliest start time for each activity;

(2)

(ii) the latest finish time for each activity.

(2)

(c) State the minimum completion time for the decorating project and identify the critical path.

(2)

(d) Activity F takes 4 days longer than first expected.

(i) Determine the new earliest start time for activities H and I.

(2)

(ii) State the minimum delay in completing the project.

(1)

(Total 12 marks)

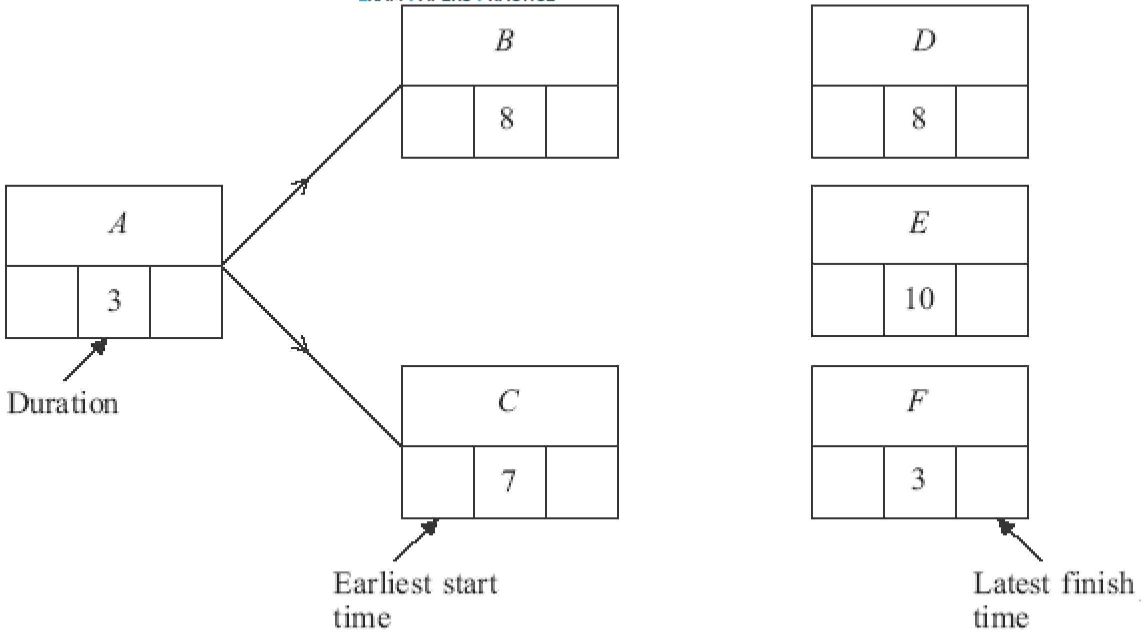
Q13.

A group of workers is involved in a building project. The table shows the activities involved. Each worker can perform any of the given activities.

Activity	Immediate predecessors	Duration (days)	Number of workers required
A	–	3	5
B	A	8	2
C	A	7	3
D	B, C	8	4
E	C	10	2
F	C	3	3
G	D, E	3	4
H	F	6	1
I	G, H	2	3

(a) Complete the activity network for the project on **Figure 1**.

Figure 1



(2)

(b) Find the earliest start time and the latest finish time for each activity, inserting their values on **Figure 1**.

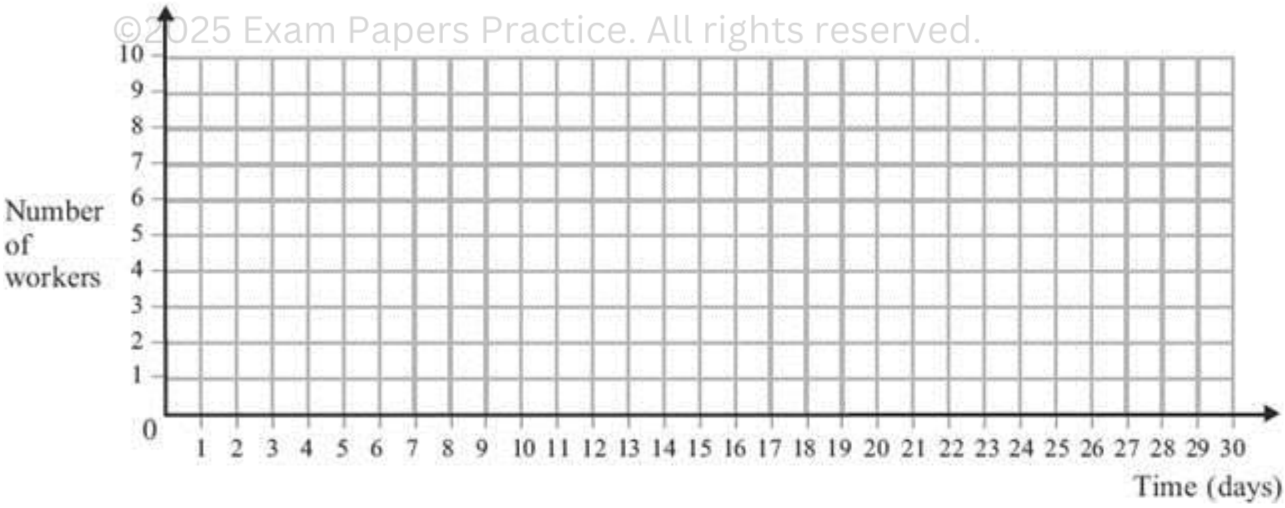
(4)

(c) Find the critical path and state the minimum time for completion.

(2)

(d) The number of workers required for each activity is given in the table above. Given that each activity starts as early as possible and assuming there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2**, indicating clearly which activities take place at any given time.

Figure 2



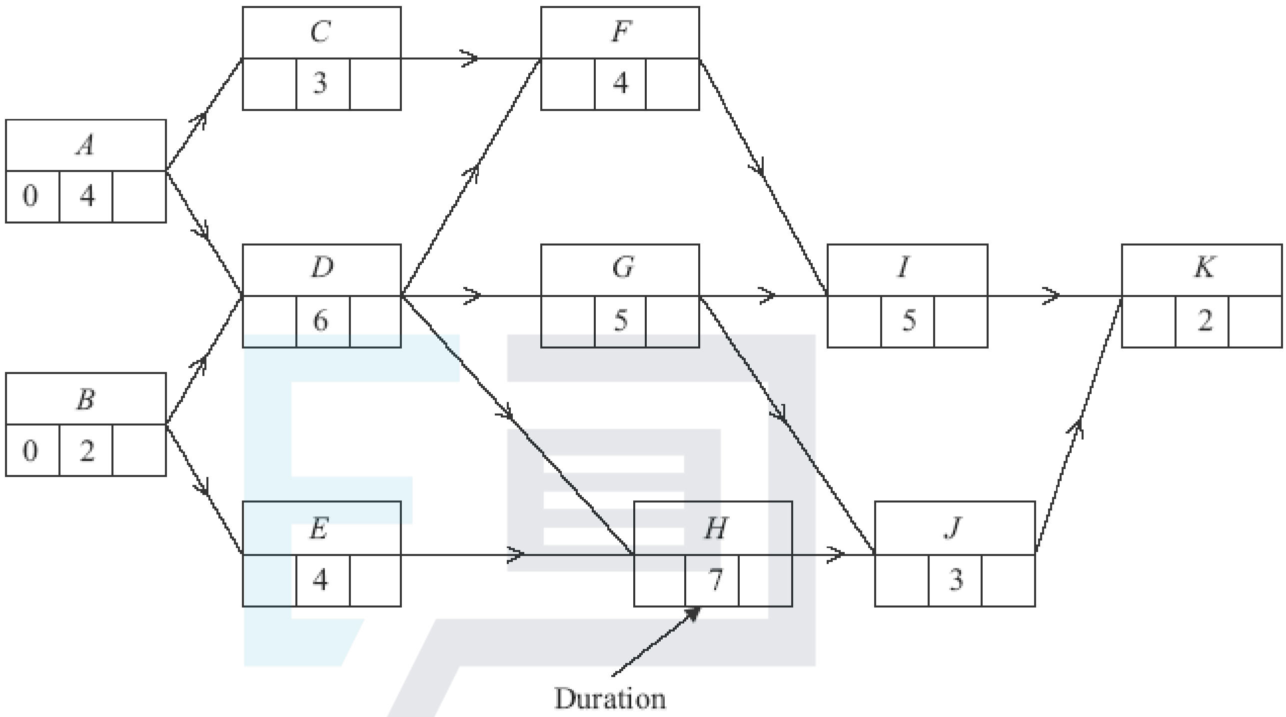
(4)

(e) It is later discovered that there are only 7 workers available at any time. Use resource levelling to explain why the project will overrun and indicate which activities need to be delayed so that the project can be completed with the minimum extra time. State the minimum extra time required.

(3)
(Total 15 marks)

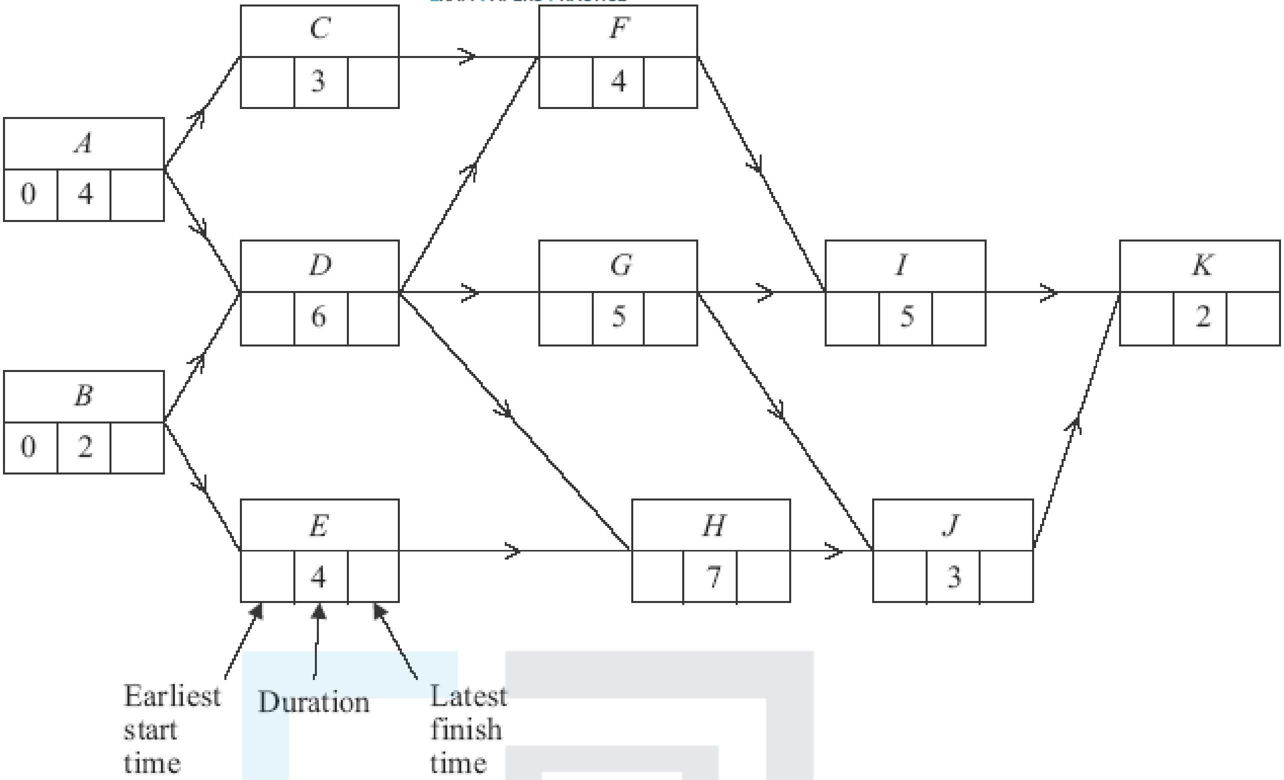
Q14.

The following diagram shows an activity network for a project. The time needed for each activity is given in days.



(a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



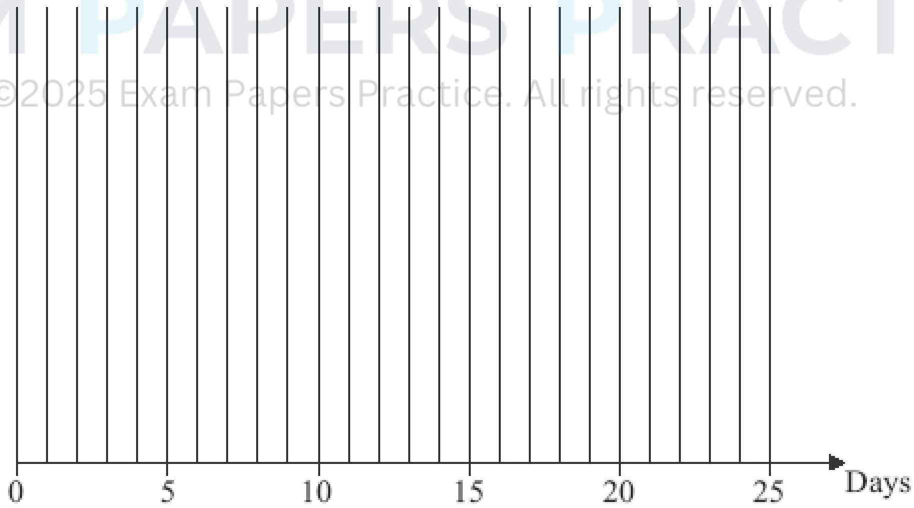
(4)

(b) Find the critical paths and state the minimum time for completion.

(3)

(c) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming each activity starts as early as possible.

Figure 2



(3)

(d) Activity C takes 5 days longer than first expected. Determine the effect on the earliest start time for other activities and the minimum completion time for the project.

(2)

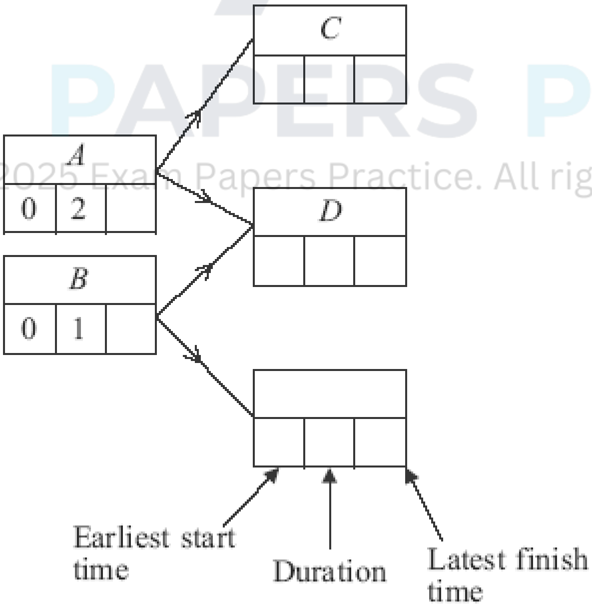
(Total 12 marks)

Q15.

A building project is to be undertaken. The table shows the activities involved.

Activity	**Immediate Predecessors**	**Duration (weeks)**
A	–	2
B	–	1
C	A	3
D	A, B	2
E	B	4
F	C	1
G	C, D, E	3
H	E	5
I	F, G	2
J	H, I	3

(a) Complete an activity network for the project on the figure below.



(3)

(b) Find the earliest start time for each activity.

(2)

(c) Find the latest finish time for each activity.

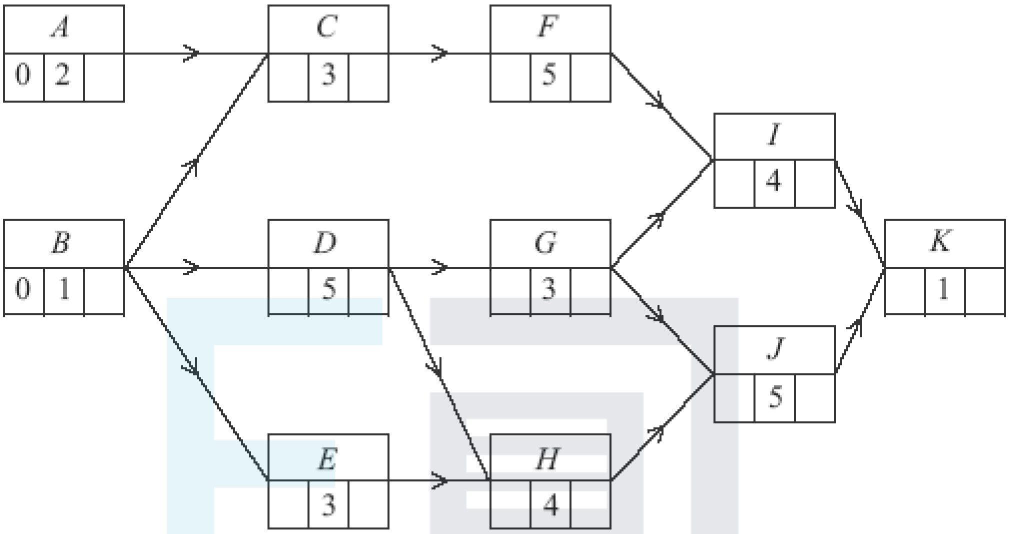
(2)

(d) State the minimum completion time for the building project and identify the critical paths.

(4)
(Total 11 marks)

Q16.

The following diagram shows an activity diagram for a building project. The time needed for each activity is given in days.

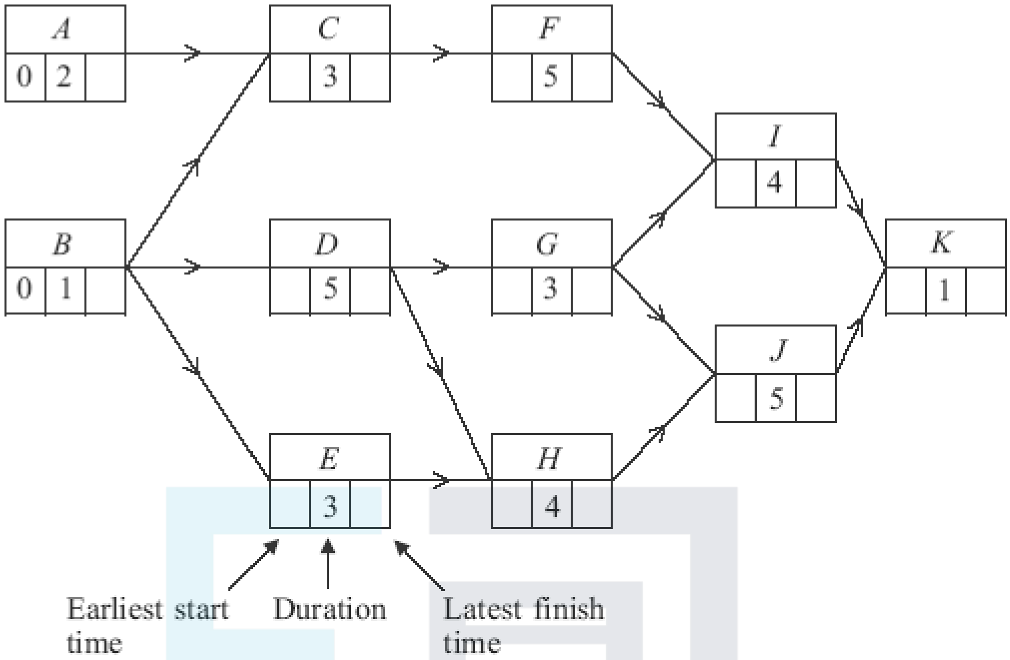


(a) Complete the precedence table below for the project.

Activity	Immediate Predecessors
A	–
B	–
C	
D	
E	
F	
G	
H	
I	
J	
K	

(2)

(b) Find the earliest start times and latest finish times for each activity and insert their values on the diagram below.



(4)

(c) Find the critical path and state the minimum time for completion of the project.

(2)

(d) Find the activity with the greatest float time and state the value of its float time.

(2)

(Total 10 marks)

Q17.

A building project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (days)	Number of Workers Required
A	–	2	3
B	*A*	4	2
C	*A*	6	1
D	*B*,*C*	8	3
E	*C*	3	2
F	*D*	2	2

G	*D, E*	4	2
H	*D, E*	6	1
I	*F, G, H*	2	3

(a) Complete the activity network for the project on **Figure 1.**

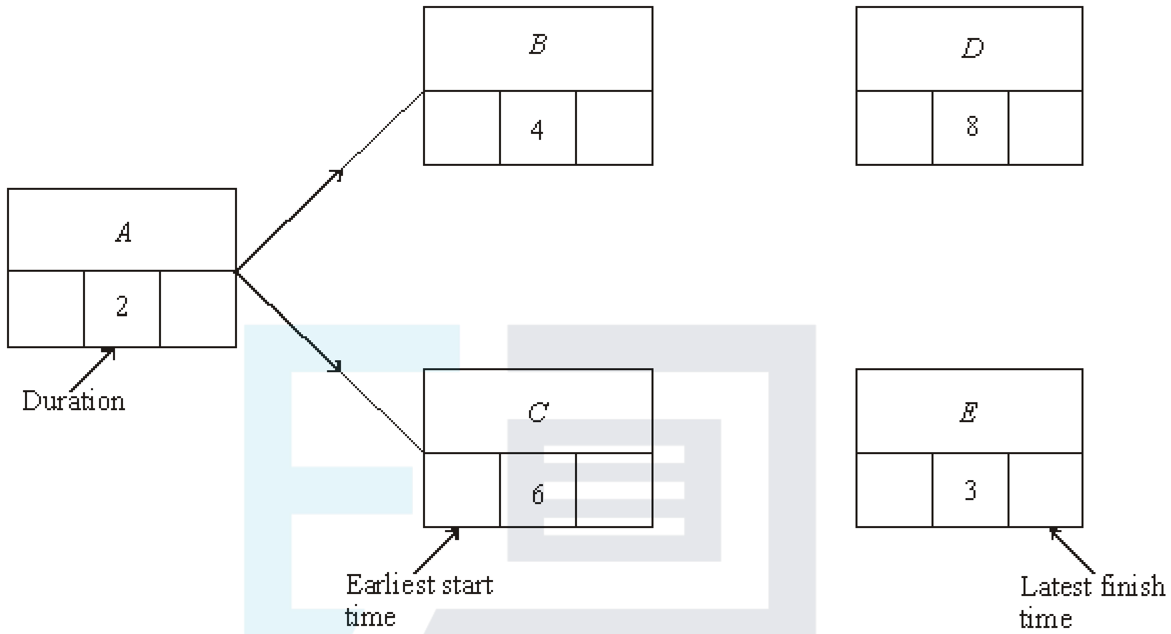


Figure 1

(3)

(b) Find the earliest start time for each activity.

(2)

(c) Find the latest finish time for each activity.

(2)

(d) Find the critical path and state the minimum time for completion.

(2)

(e) State the float time for each non-critical activity.

(2)

(f) Given that each activity starts as early as possible, draw a resource histogram for the project on **Figure 2.**

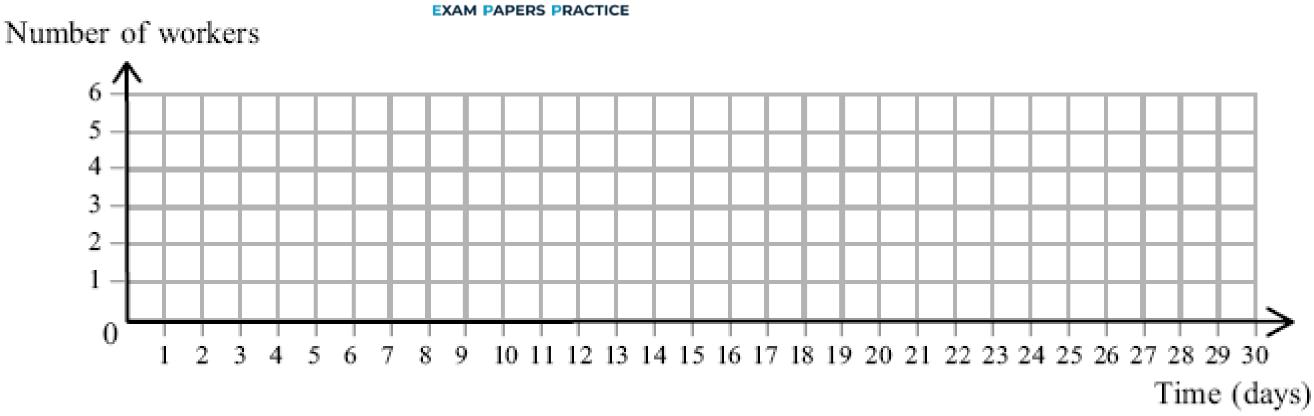


Figure 2

(4)

(g) There are only 3 workers available at any time. Use resource levelling to explain why the project will overran and state the minimum extra time required.

(3)

(Total 18 marks)

Q18.

[Figures 1 and 2, below, are provided for use in this question.]

Figure 1

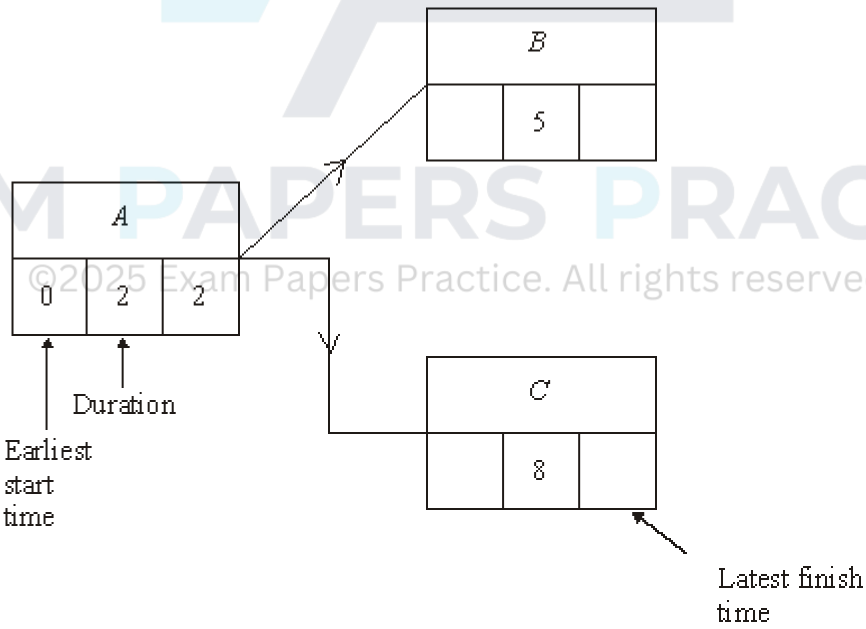
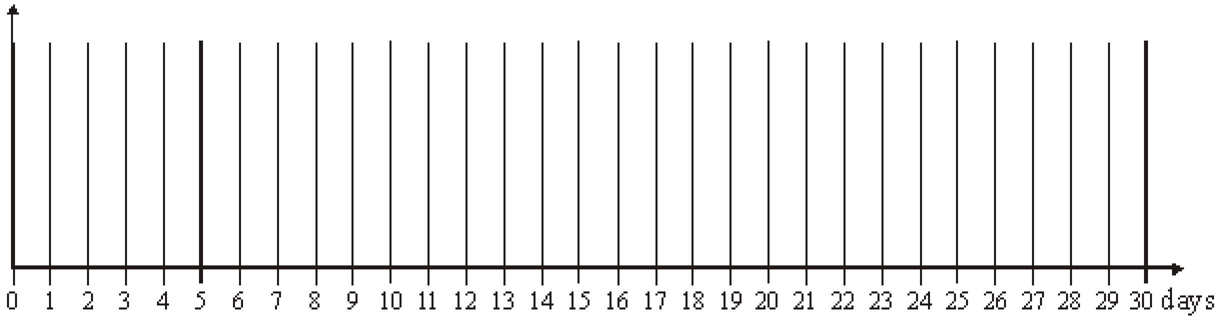


Figure 2



A construction project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (days)
A	–	2
B	A	5
C	A	8
D	B	8
E	B	10
F	B	4
G	C,F	7
H	D,E	4
I	G,H	3

(a) Complete the activity network for the project on **Figure 1.**

(3)

(b) Find the earliest start time for each activity.

(2)

(c) Find the latest finish time for each activity.

(2)

(d) Find the critical path.

(1)

(e) State the float time for each non-critical activity.

(2)

(f) **On Figure 2,** draw a cascade diagram (Gantt chart) for the project, assuming each activity starts as **late** as possible.

(4)

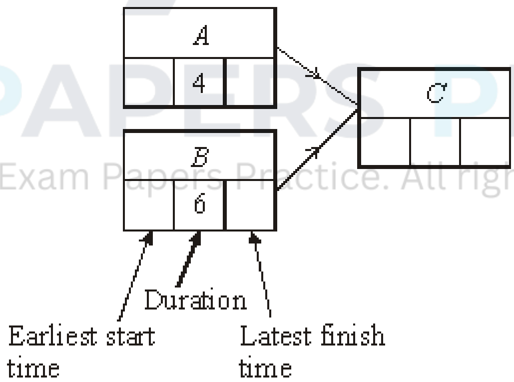
(Total 14 marks)

Q19.

A cleaning project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (hours)
A	–	4
B	–	6
C	A,B	7
D	C	9
E	C	10
F	B	3
G	D,E	6
H	F,G	5
I	G	3
J	H,I	2

(a) Complete an activity network for the project on the figure below.



(4)

(b) Find the earliest start time for each activity.

(2)

(c) Find the latest finish time for each activity.

(2)

(d) (i) Write down the shortest time in which the whole cleaning project can be completed.

(1)

(ii) State which activities are the critical ones and explain how you can recognise them.

(3)

(e) A problem occurs which means that activity I cannot be started until 3 hours after activity H has started.

(i) Find the new earliest start time for activity I, giving a reason for your answer.

(2)

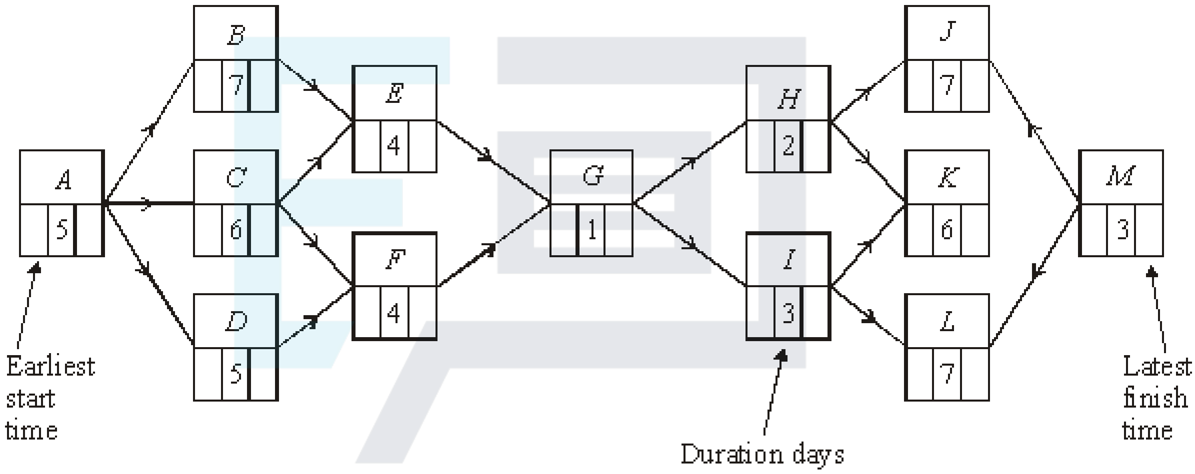
(ii) Find the revised shortest completion time for the project.

(2)

(Total 16 marks)

Q20.

The following diagram shows an activity network for a major project.



(a) On the figure above:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path.

(1)

(c) Write down the activity that has a float time of more than 1 day.

(1)

(d) Given that activity C overruns by 3 days, find the overrun time for the whole project.

(2)

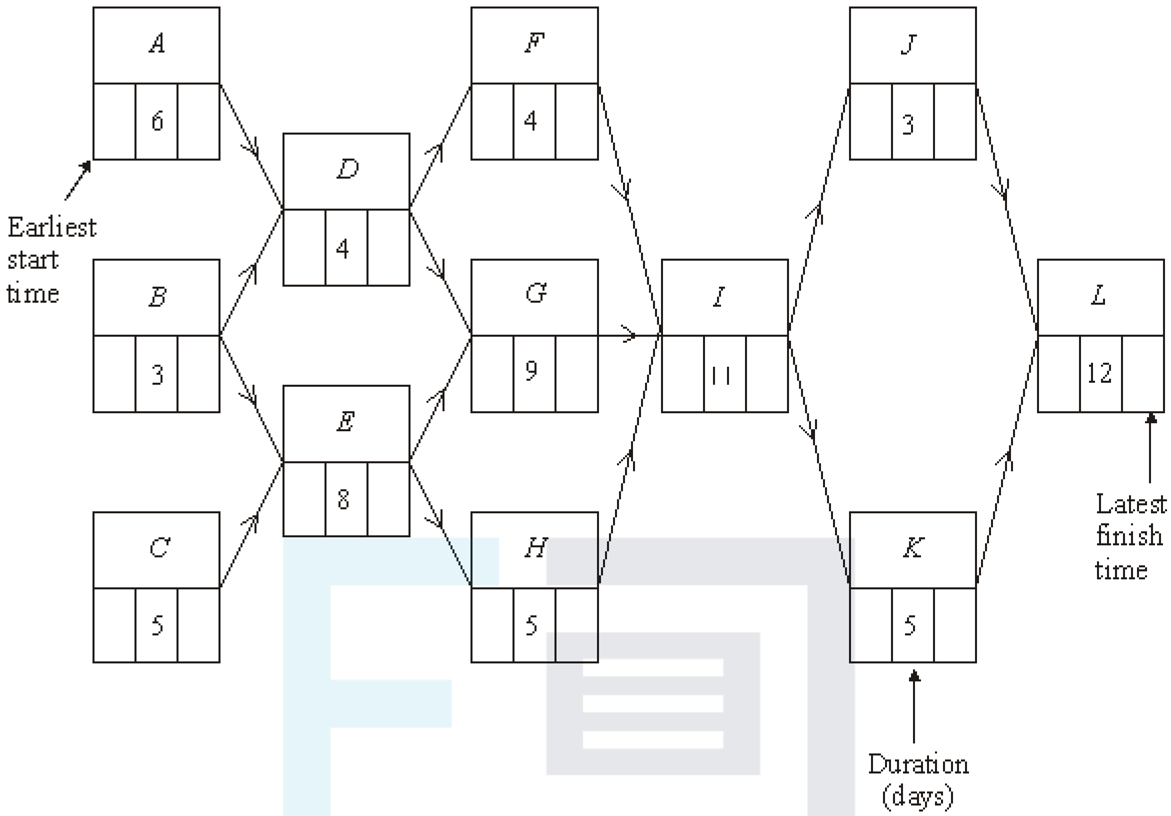
(e) Given that activities C and F overrun by 3 days each, find the overrun time for the whole project.

(3)

(Total 11 marks)

Q21.

The following diagram shows an activity network for a major building project.



(a) On the figure above find:

(i) the earliest start time for each activity;
(2)

(ii) the latest finish time for each activity.
(2)

(b) Identify the critical path.
(1)

(c) State the activity with the greatest float time.
(1)

(d) Construct a Gantt (cascade) diagram for the project.
(3)

(e) Activities D and J overrun by 5 days each.

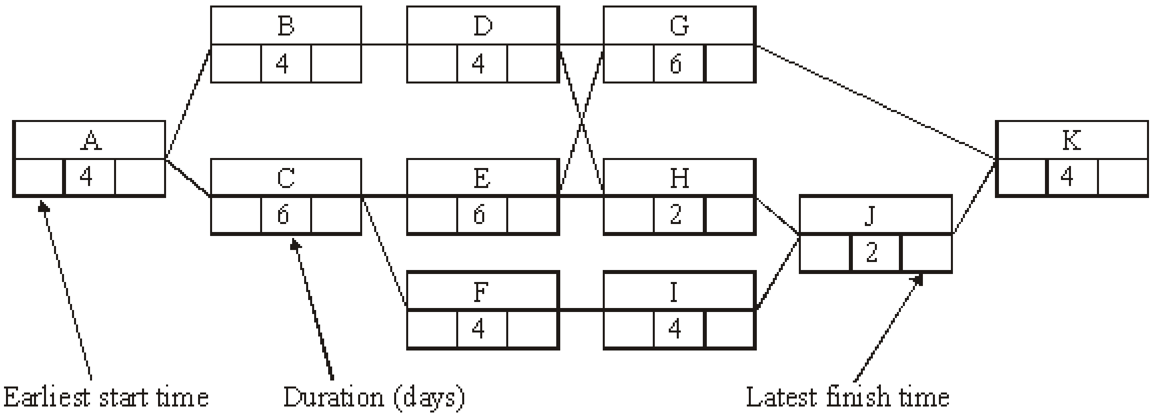
Find:

(i) the new minimum completion time for the project;
(3)

(ii) the new critical path.
(1)
(Total 13 marks)

Q22.

The diagram shows an activity network for a building project.



(a) On the diagram:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path and state the minimum time for completion of the project.

(1)

(c) Given that one activity overruns by four days, but the completion time is unaffected, list the activities which could overrun.

(1)

(d) Each activity requires one worker in order for the activity to be completed in the stated time.

Draw a resource histogram for the project.

(3)

(e) Given that there are only two workers available, each being able to complete any of the activities they undertake in the stated time:

(i) state the new minimum time to complete the project, giving a reason for your answer;

(2)

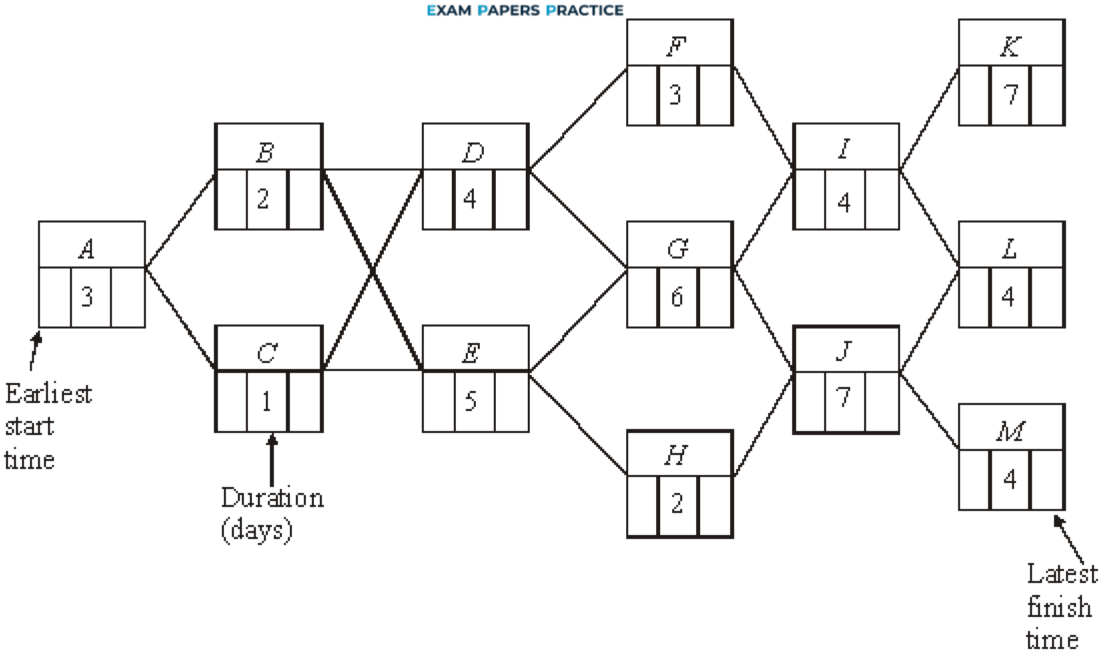
(ii) state how the activities could be divided between the two workers to achieve this time.

(1)

(Total 12 marks)

Q23.

A major construction project is to be undertaken and the project has been divided into 13 activities. The following diagram shows the activity network for the project.



(a) On the diagram above:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) List the non-critical activities.

(1)

(c) Draw a Gantt (cascade) diagram for the project.

(3)

(d) There are only two workers available, each able to complete any of the activities in the stated time.

(i) Find the new minimum completion time for the whole project.

(2)

(ii) Allocate the activities to the two workers so that this new minimum time can be achieved.

(2)

(Total 12 marks)

Q24.

A building company is going to lay a new block-paved driveway. The table below shows the activities involved.

	Activity	Immediate Predecessor	Duration (days)
A	Remove tarmac top drive	—	7

B	Remove tarmac bottom drive	—	6
C	Concrete edging top	A	3
D	Concrete edging bottom	B	3
E	Lay hardcore	C, D	2
F	Lay sand	E	2
G	Lay blocks	F	5
H	Brush sand	G	1

(a) Construct an activity network for the project.

(3)

(b) Find the minimum completion time for the whole project.

(2)

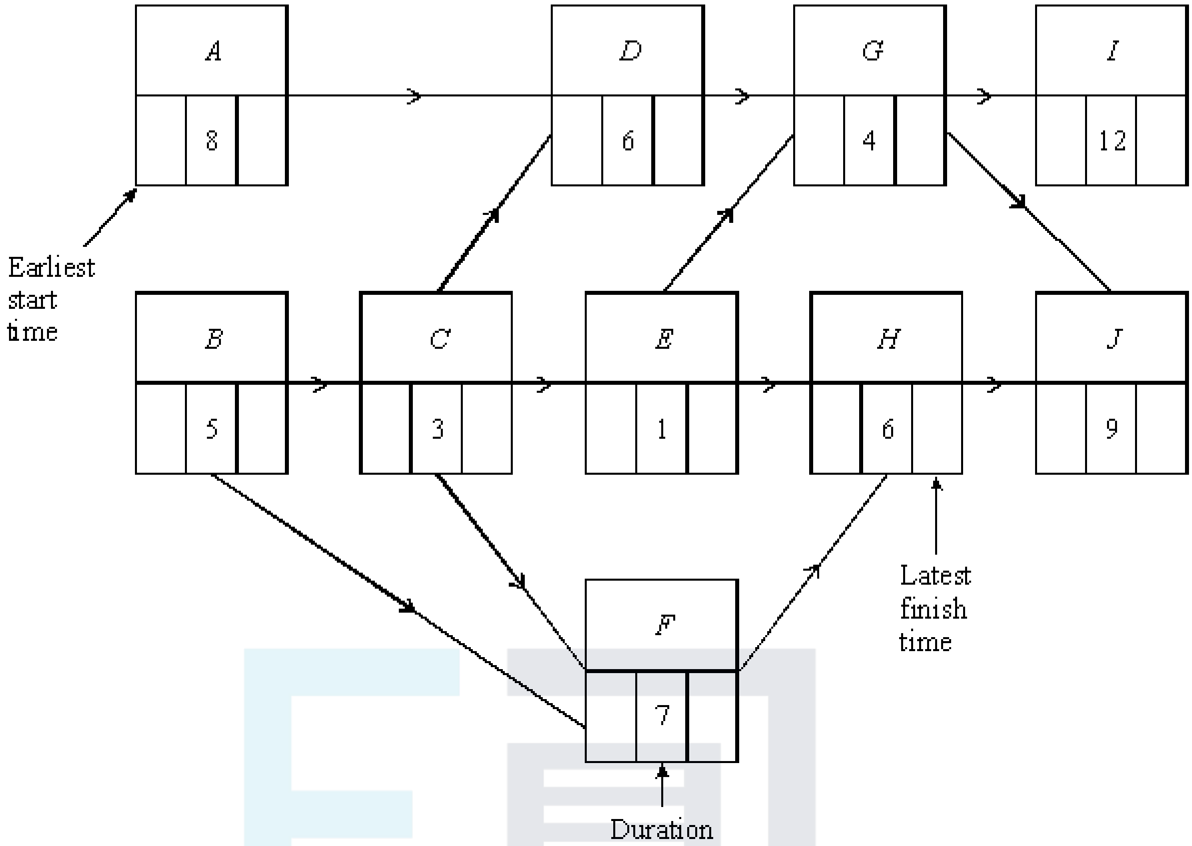
(c) Given that each activity starts as early as possible, draw a cascade (Gantt) chart for the project.

(4)

(Total 9 marks)

Q25.

The following diagram shows an activity network for a project. The time required for each activity is given in hours.



(a) On the table below, complete the precedence table.

Activity	Immediate Predecessor
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	

(2)

(b) On the diagram above, complete the earliest start and latest finish times for each activity.

(4)

(c) List the critical activities.

(1)

(d) Draw a Gantt diagram, given that the project is to be completed in the minimum possible time and all the activities are to start as late as possible.

(4)

(c) There are only two workers available. They are both capable of completing each activity on their own in the stated time.

Find the minimum overrun time for the whole project.

(2)
(Total 13 marks)

Q26.

[Graph paper is provided for use in answering this question.]

A small building project is to be undertaken. The following precedence table shows each activity, its duration, and the number of workers required to complete the activity.

Activity	Immediate predecessor	Duration (days)	Number of workers
A	-	3	4
B	A	4	2
C	A	2	1
D	B	3	2
E	D	11	4
F	D	4	2
G	C, D	5	2
H	F, G	2	1
I	E, H	2	4

(a) Construct an activity network for the project.

(3)

(b) Find the earliest start time for each activity.

(2)

(c) Find the latest finish time for each activity.

(2)

(d) State the float time for each non-critical activity.

(2)

(e) Given that each activity starts as early as possible, draw a resource histogram for the project.

(4)
(Total 13 marks)

Q27.

A major construction project is to be undertaken and the project has been divided into 12 activities. The following diagram shows the activity network for the project.

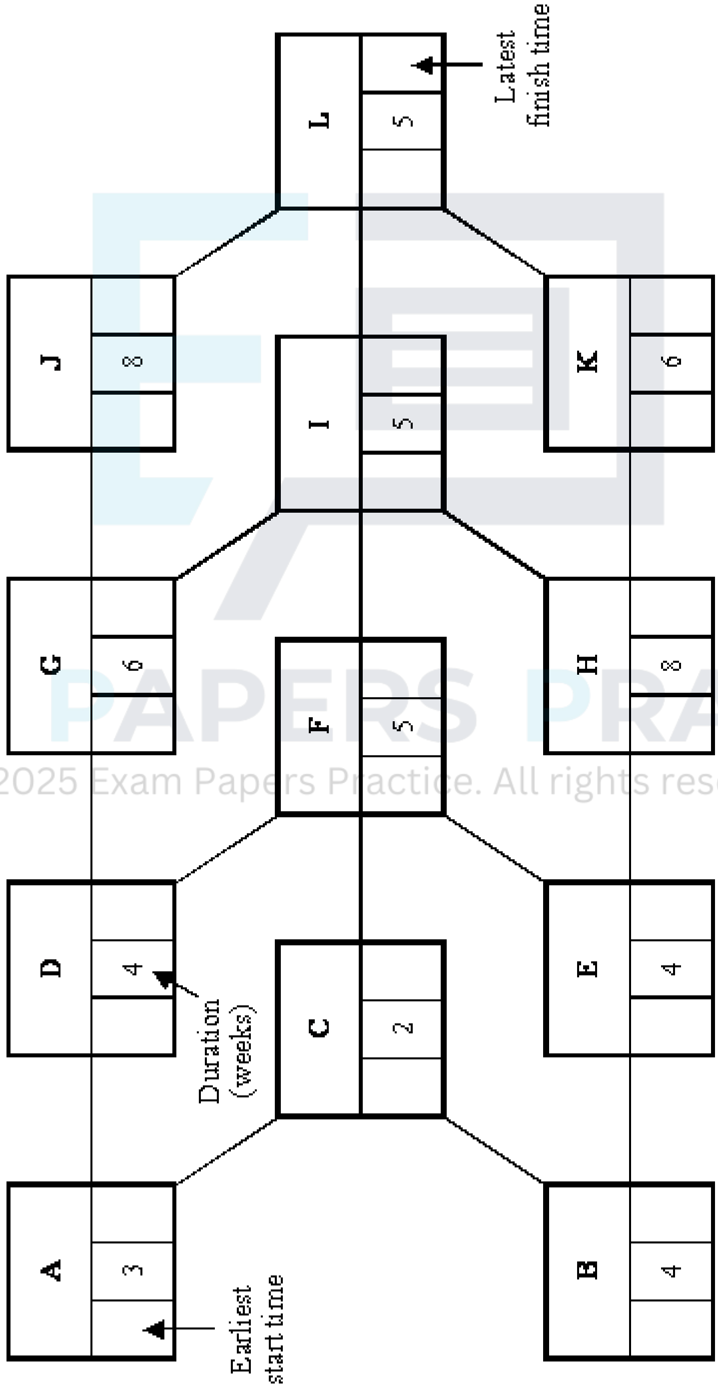


Figure 1

(a) On **Figure 1**:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path and state the shortest completion time for the whole project.

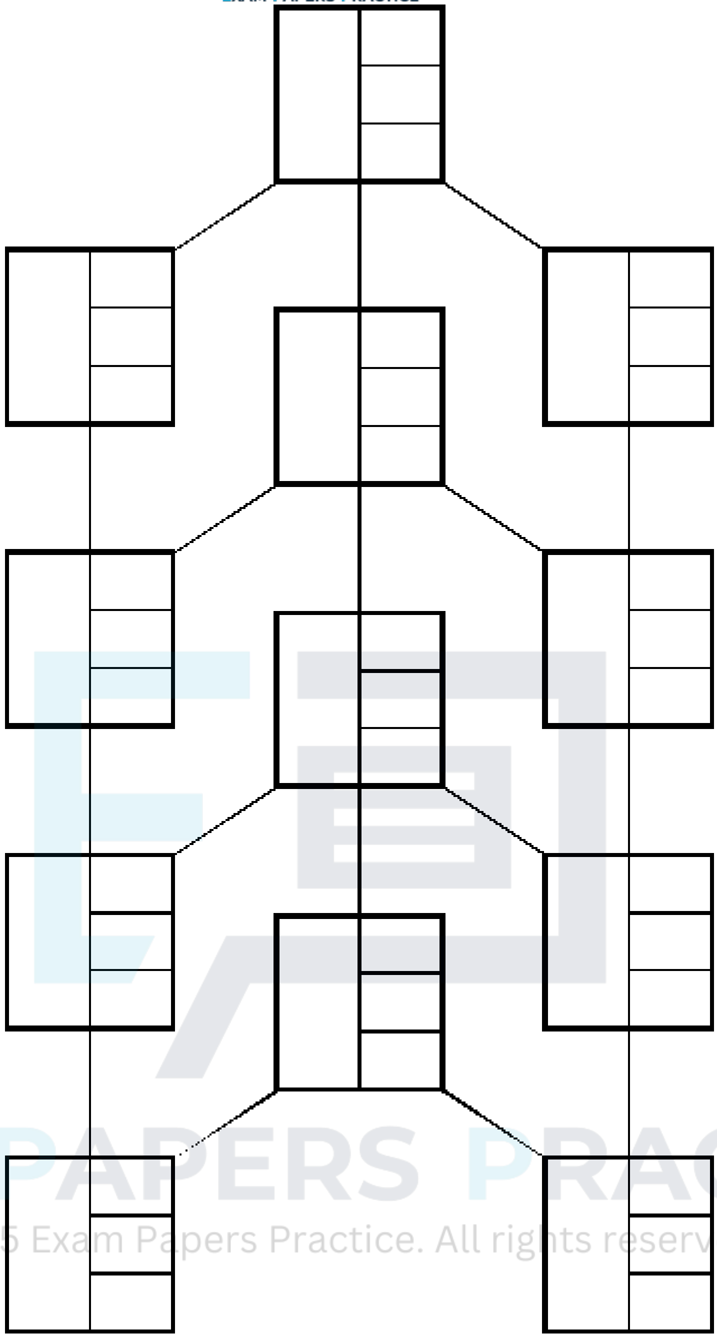
(1)

Some of the activities can be speeded up at an additional cost. The following table lists the activities that can be speeded up, together with the maximum time reduction in each case, and the extra cost for each week of reduction, in pounds per week.

Activity	**Maximum time reduction** (weeks)	**Extra cost for each week of reduction** (pounds per week)
A	1	12 000
B	2	20 000
C	1	10 000
F	1	14 000
J	3	15 000
K	4	24 000

The company wants to complete the project as soon as possible.

(c) (i) Find the revised minimum time for the completion of the whole project.
[You may use **Figure 2**, to answer this question.]



(2)

(ii) For **each** of the six activities that can be speeded up, state with justification, whether the activity should be speeded up and the reduction in the number of weeks.

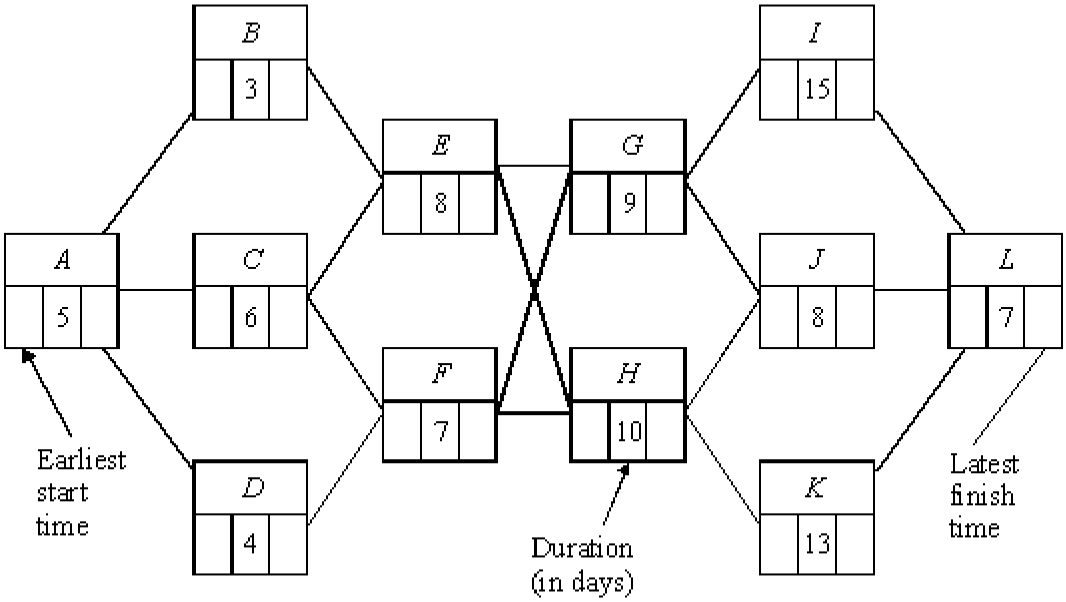
(5)

(iii) Calculate the total extra cost the company would incur in meeting this revised completion time.

(1)
(Total 13 marks)

Q28.

The following diagram shows an activity network for a major building programme.



(a) On diagram,

(i) find the earliest start time for each activity,

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path.

(1)

(c) Construct a Gantt diagram for the project.

(3)

(d) Given that at most two activities can take place at the same time, find the new minimum completion time for the whole project

(3)

(Total 11 marks)

Q29.

The shell of a flat has been constructed and the inside is to be completed. The work has been divided into 13 activities, as shown in the table below.

Activity		Tradesman	Immediate predecessors	Duration (days)
A	Studding	Builder		2
B	Electrics (first fix)	Electrician	A	1
C	Plumbing (first fix)	Plumber	A	1.5
D	Plaster boarding	Plasterer	B, C	1

E	Plastering	Plasterer	D	1
F	Artexing	Decorator	E	1
G	Electrics (second fix)	Electrician	E	0.5
H	Joinery (first fix)	Joiner	E	1
I	Plumbing (second fix)	Plumber	E	1
J	Joinery (second fix)	Joiner	G, H, I	2
K	Painting	Decorator	F, J	1
L	Tiling	Tiler	I, J	0.5
M	Cleaning	Cleaner	K, L	0.5

(a) Construct an activity network for the project.

(4)

(b) Find the earliest start time for each activity.

(3)

(c) Find the latest finish time for each activity.

(3)

(d) Identify the critical activities and state the shortest completion time for the whole project.

(2)

(e) Construct a cascade (Gantt) diagram for the project, assuming that each activity is to start as early as possible.

(3)

(f) The foreman of the project can speed up the project by using either an extra electrician, or an extra plumber or an extra joiner.

An extra electrician would reduce the time for electrical work by 50%, an extra plumber would reduce the time for plumbing work by 25%, and an extra joiner would reduce the time for joinery work by 20%.

Explain which extra tradesman the foreman should use, and find the new minimum time to complete the project.

(5)

(Total 20 marks)

Q30.

A group of workers are involved in a self-build project. The project has been divided into the activities shown in the table below.

Activity	Immediate predecessors	Duration (days)	Number of workers required
A	–	2	8
B	A	6	4
C	A	12	4
D	B	12	8
E	B	16	4
F	B	4	6
G	C, F	10	6
H	E	4	4
I	D, G, H	2	8

(a) Assuming that sufficient workers are available,

(i) construct an activity network for the project,

(4)

(ii) find the earliest start time for each activity,

(3)

(iii) find the latest finish time for each activity,

(3)

(iv) identify the critical path and state the shortest completion time for the whole project.

(2)

(b) Construct a resource histogram for the project showing the number of workers required each day, assuming each activity is to start as early as possible.

(4)

(c) Given that the group consists of 18 workers, explain why it would be necessary to reschedule some of the activities to enable them to finish the project in the shortest possible completion time. Give **two** possible ways in which this could be done.

(4)

(Total 20 marks)

Q31.

A building project has been divided into a number of activities as shown in the table below.

	Activity	Immediate	Duration

		predecessor	(days)
A	Prepare site	–	1
B	Erect walls	A	4
C	Excavate and lay drains	A	7
D	Fit window frames	B	3
E	Erect roof	B	9
F	Install electrics	B	7
G	Install plumbing	C, D	6
H	Insulate roof	E	3
I	Plaster walls	F, G, H	2

(a) Construct an activity network for the project.

(4)

(b) Find the earliest start time for each activity.`

(3)

(c) Find the latest finish time for each activity.

(3)

(d) Identify the critical path and state the shortest completion time for the whole project.

(2)

(e) Construct a cascade (Gantt) diagram for the project, assuming each activity is to start as early as possible.

(4)

(f) Each activity requires one worker. Draw a resource histogram showing the number of workers requires each day

(2)

(g) Given that there are only three workers available on any one day, show how the activities could be allocated so that the project may be finished in the shortest completion time.

(2)

(Total 20 marks)

Q32.

A major construction project is to be undertaken and the project has been divided into 11 activities, as shown in the table below.

Activity	Immediate predecessors	Duration

		(weeks)
A	–	3
B	–	4
C	A	2
D	B	3
E	C, D	5
F	C	6
G	D	7
H	E, F, G	4
I	G	6
J	F	8
K	H, I, J	3

(a) Construct an activity network for the project.

(4)

(b) Find the earliest start time for each activity.

(3)

(c) Find the latest finish time for each activity.

(3)

(d) Identify the critical path and state the shortest completion time for the whole project.

(2)

Some of the activities can be speeded up at an additional cost. The following table lists the activities that can be speeded up, their additional cost in £ per week, together with the minimum times required to complete the activities.

Activity	Additional cost (£ **per week)**	Minimum time (weeks)
E	5000	1
I	4000	4
J	3000	5

The company wants to complete the project as soon as possible.

(e) (i) Find which activities should be speeded up. For **each** such activity, state, with justification, the reduction in the number of weeks.

(4)

(ii) Hence state the revised minimum time for the completion of the whole project.

(1)

(iii) Calculate the total additional cost the company would incur in meeting this revised completion time.

(2)
(Total 19 marks)

Q33.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

Each year the secretary of a Mathematical Society undertakes the task of preparing and distributing the society's annual programme. The programme gives dates of meetings, a list of speakers and the titles of their talks, and an up-to-date list of paid-up members. The table below shows the activities which the secretary needs to carry out in preparing and distributing the programme.

	Activity	Immediate predecessor	Duration (days)
A	Select dates for meetings	–	3
B	Book speakers	A	13
C	Obtain advertising material for programme	A	12
D	Send out membership renewal notices	–	21
E	Prepare list of paid–up members	D	6
F	Prepare programme for printers and check proofs	B, C, E	10
G	Assemble material for circulation	F	7
H	Print members' address labels	E	4
I	Send out programmes	G H	3

(a) Construct an activity network for the project.

(4)

(b) On your network show full details of the earliest and latest starting and finishing times for each activity. Indicate and state the critical path.

(6)

(c) For each activity state the float time.

(2)

(d) Construct a cascade diagram for the project.
If each activity requires one helper so that activities are completed on time, state the minimum number of helpers required for the project.

(5)

(e) The secretary is let down by one helper who resigns immediately prior to the start of the project and so cannot be replaced. Find the least possible number of days' delay to the completion of the project and explain how it can be achieved.

(2)
(Total 19 marks)

Q34.

A garden is to be completely renovated with the project being broken down into the activities shown in the table below.

Activity	Description	Duration (days)	Immediate Predecessors
A	Clear site	2	-
B	Prepare base for summer house	2	A
C	Prepare base for pergola	3	A
D	Erect summer house	2	B
E	Erect pergola	3	C
F	Prepare patio	2	C
G	Lay stepping stones	3	D, E
H	Build rockery	1	D, E
I	Lay patio	2	F
J	Plant out	1	H, I
K	Tidy site	2	G, J

(a) Construct an activity network for the project. **(4)**

(b) Find the earliest start time for each event. **(3)**

(c) Find the latest finish time for each event. **(3)**

(d) Identify the critical path and state the shortest completion time for the whole project. **(2)**

(e) If one of the activities overruns by two days, but the completion time is unaffected, list the activities which could overrun. **(2)**

(f) If only two workers were available, each being completely responsible for the activities they undertake,

(i) state the minimum time to complete the project, giving a reason for your answer; **(2)**

(ii) state how the activities could be divided between the two workers to achieve this time.

(1)
(Total 17 marks)

Q35.

The table shows the activities involved in a project, their durations, and their immediate predecessors.

Activity	A	B	C	D	E	F	G	H
Duration (days)	3	1	1	4	5	2	2	2
Immediate predecessors	–	–	B	A	A, C	B	F	D, E

(a) Draw an activity network for the project.

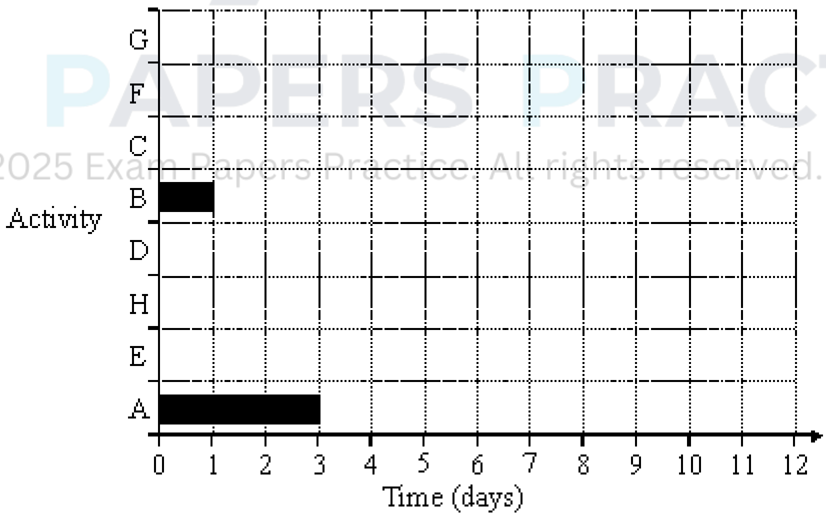
(5)

(b) Perform a forward pass and a backward pass on your activity network. Give the minimum completion time and the activities forming the critical path.

(6)

(c) In a cascade chart each activity is represented by a bar showing when it is scheduled. In the chart below, activities A and B are shown, both starting as early as possible.

Copy and complete the cascade chart, with the activities ordered as shown, and with each activity starting as early as possible.



(3)

(d) The number of people needed for each activity is as follows:

Activity	A	B	C	D	E	F	G	H
People	3	2	2	2	3	1	1	4

Activities are to be scheduled, some later than their earliest possible start time, so that at most 5 people are needed at any one time, whilst the project is still completed in the minimum time.

What are the new scheduled start times for activities D, F and G?

(2)

(Total 16 marks)